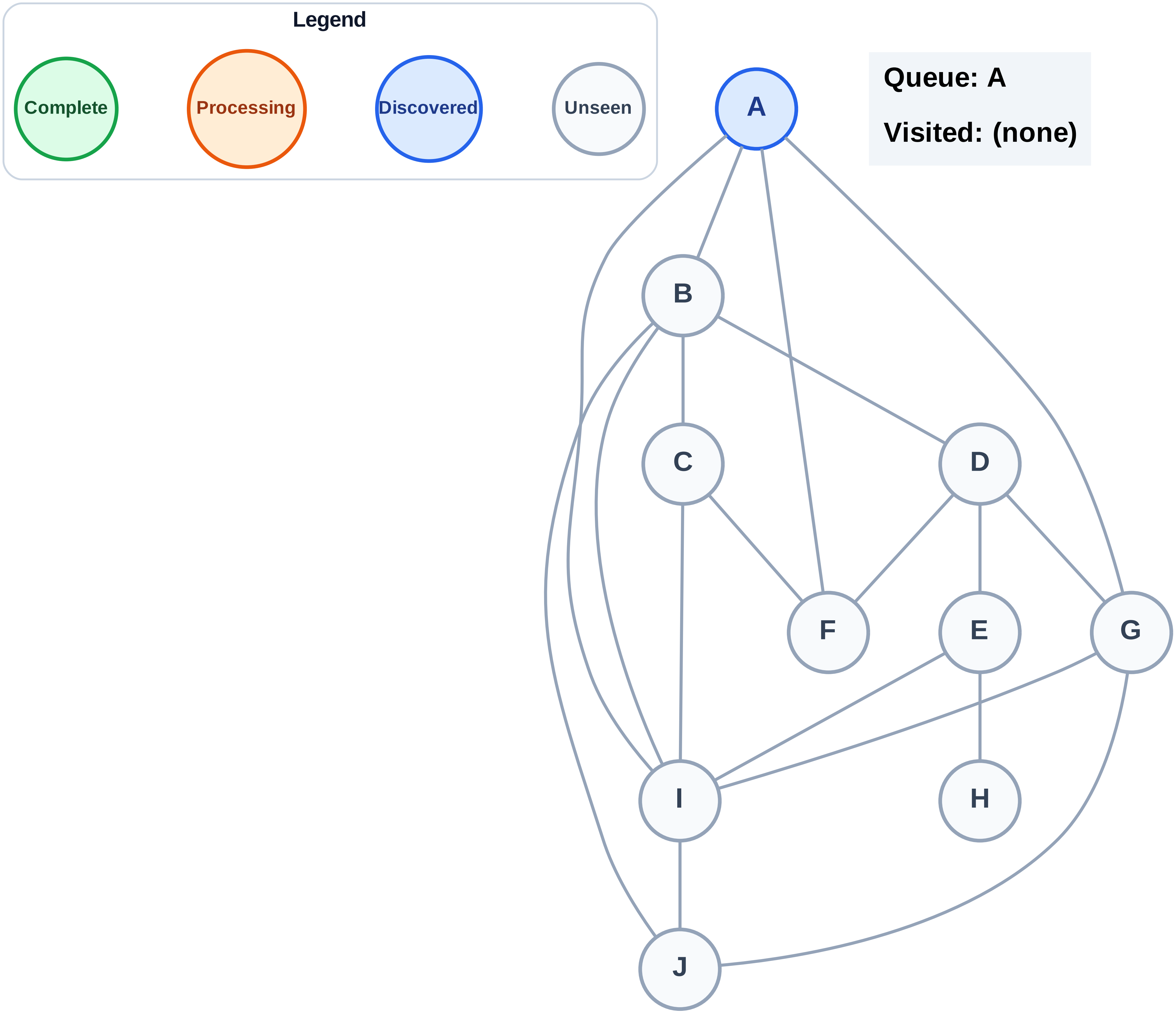
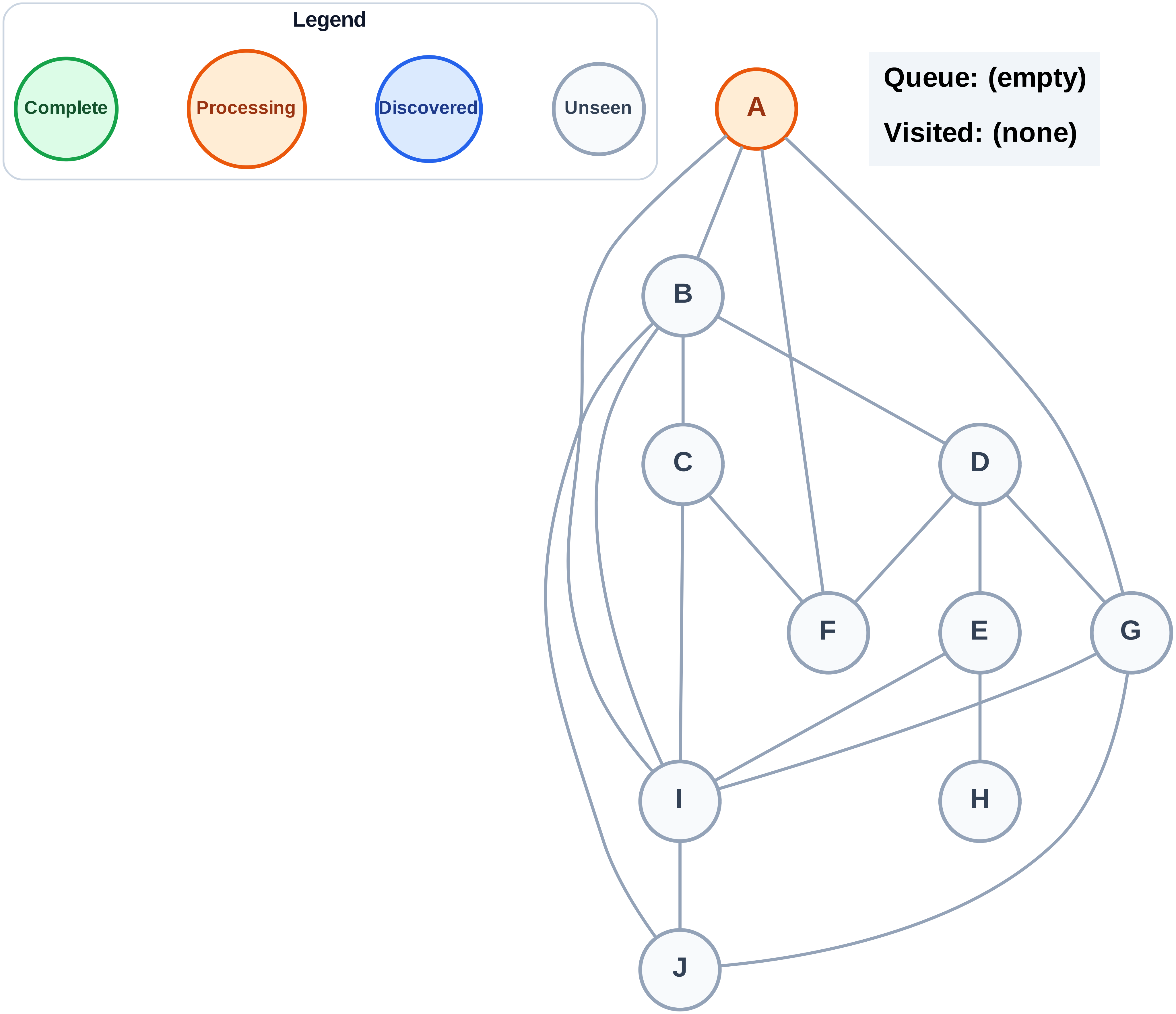
BFS Traversal

Step 1: Start at A



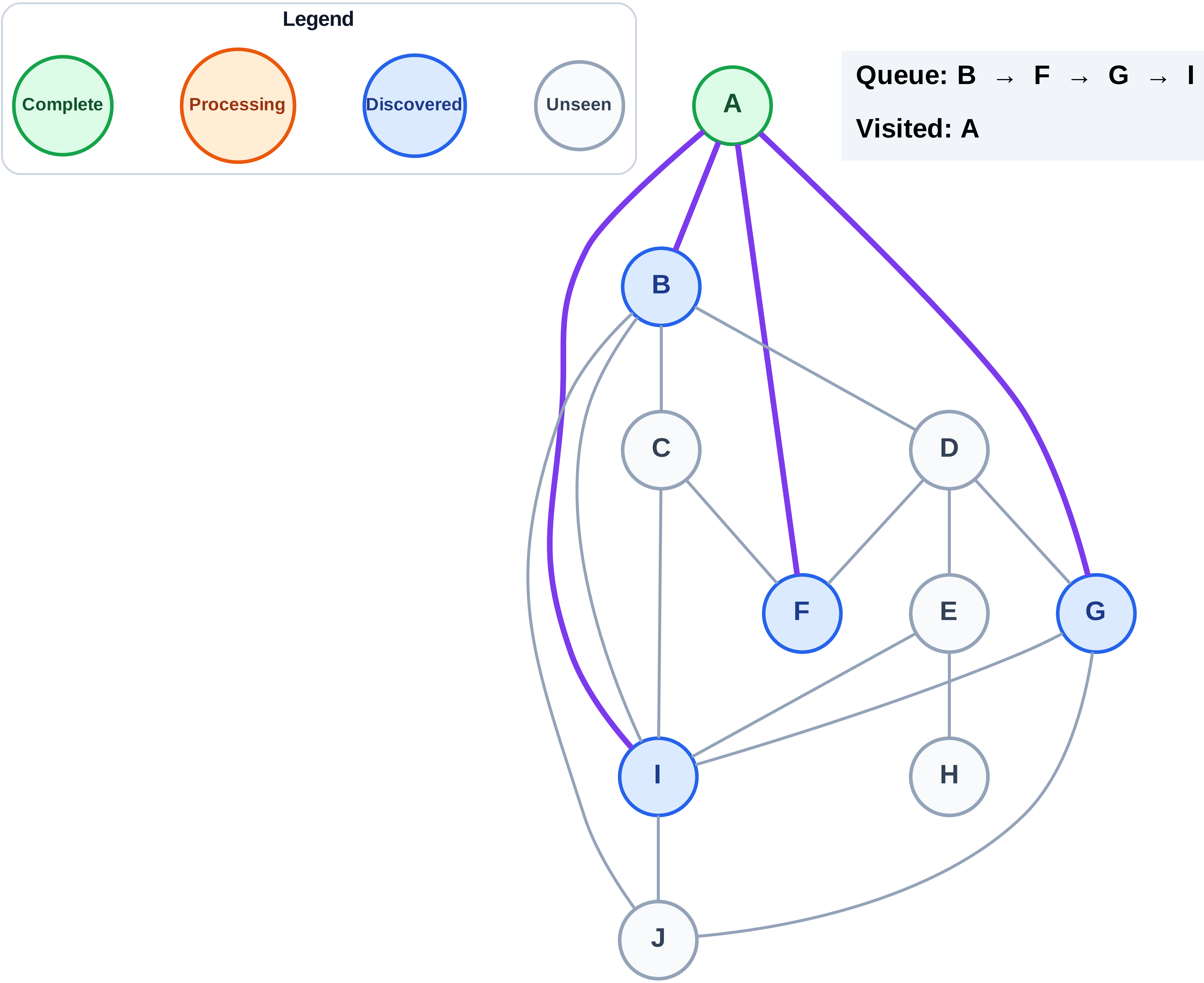
BFS Traversal

Step 2: Process node A



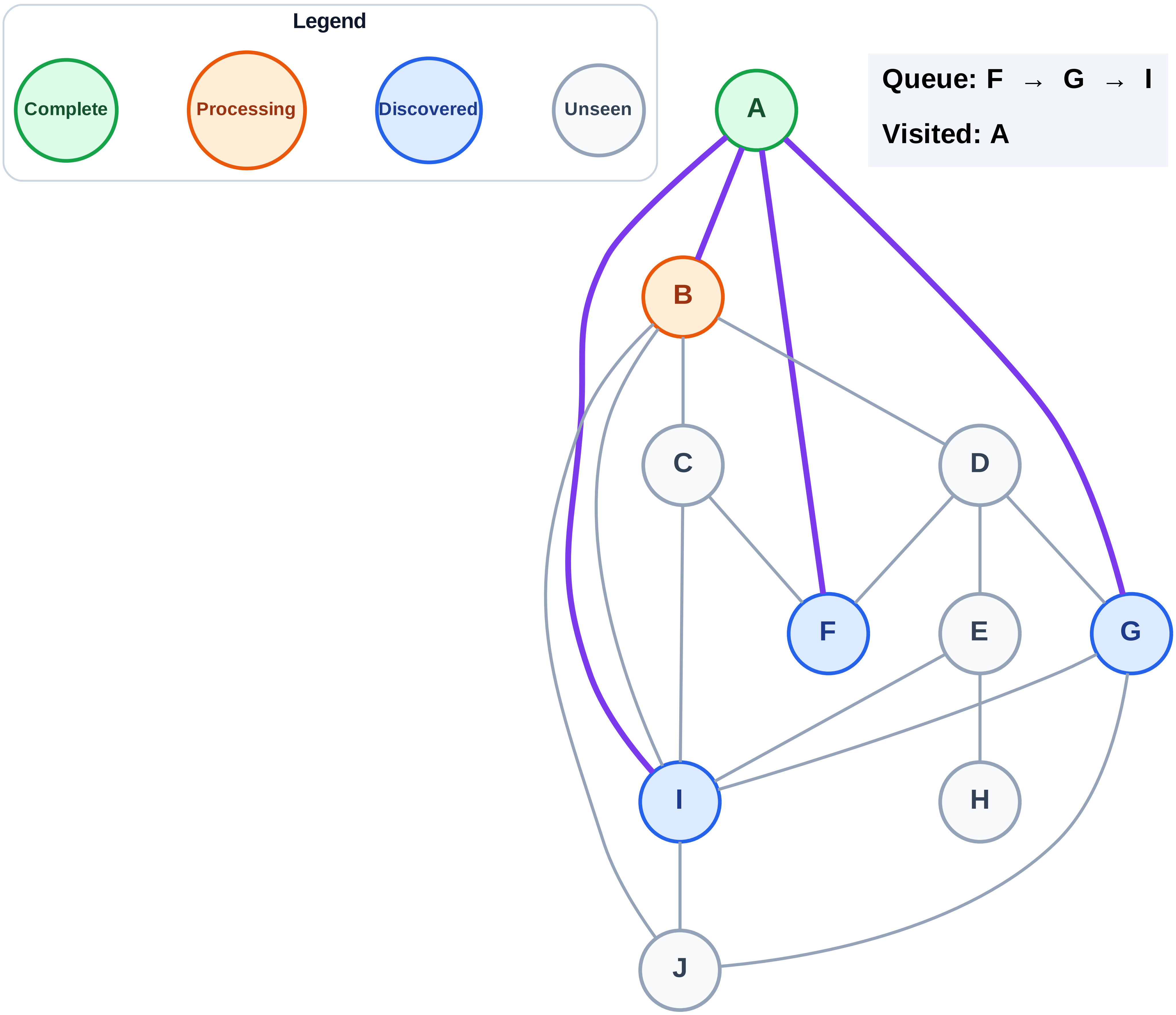
BFS Traversal

Step 3: Discover B, F, G, I from A



BFS Traversal

Step 4: Process node B

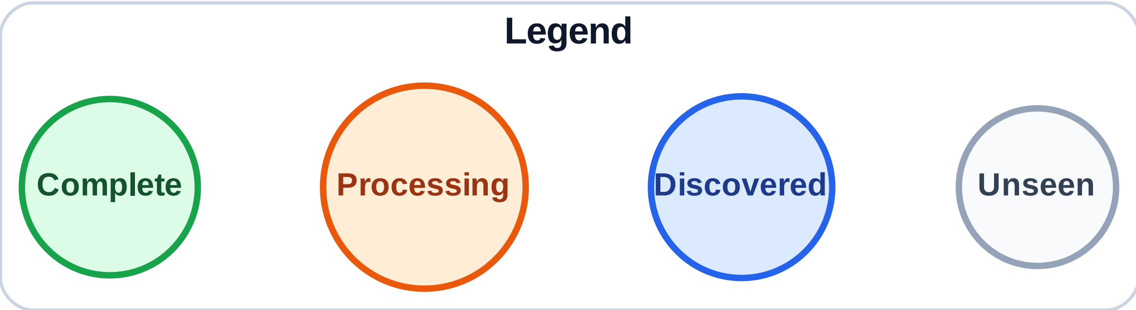


Queue: F → G → I

Visited: A

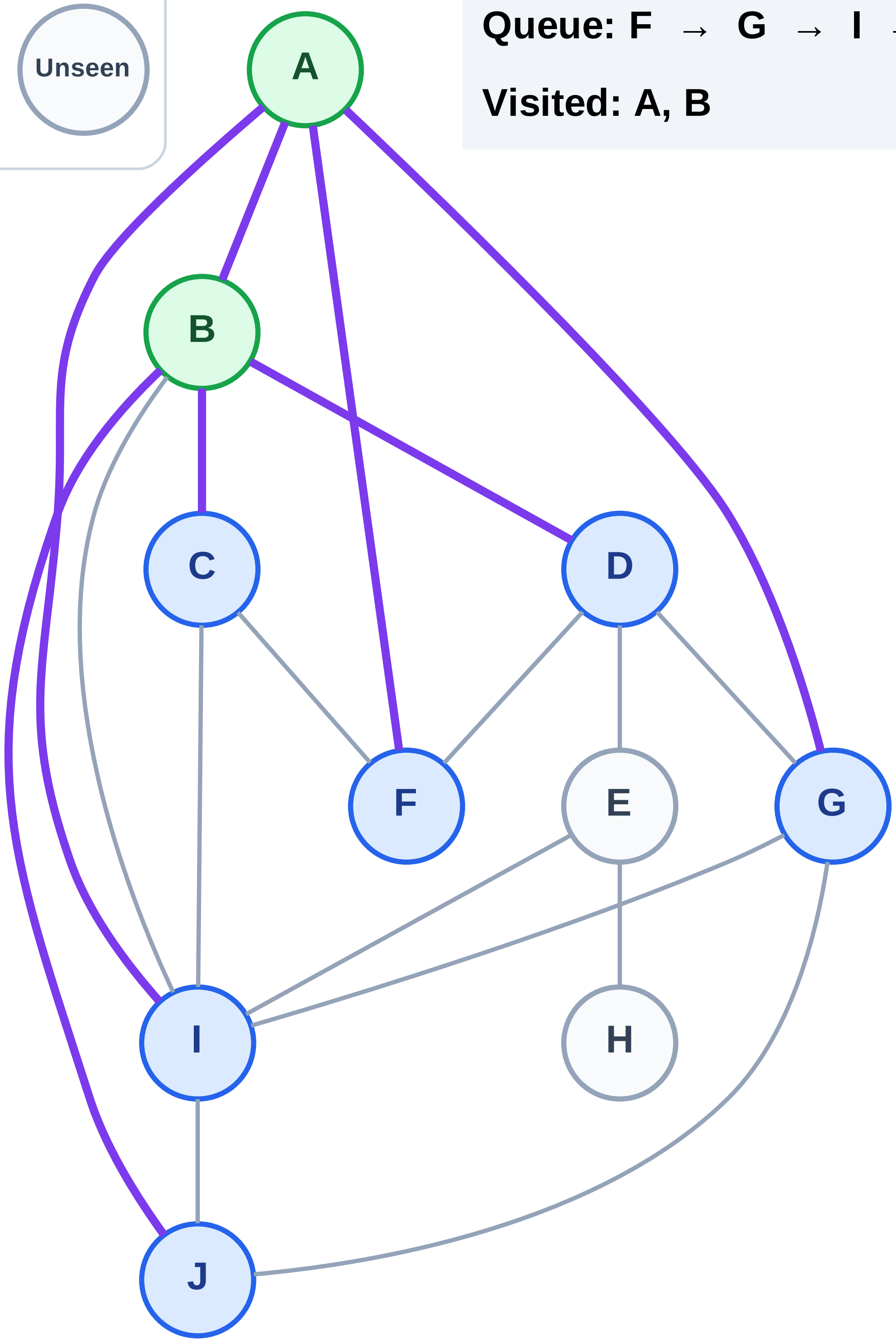
BFS Traversal

Step 5: Discover C, D, J from B



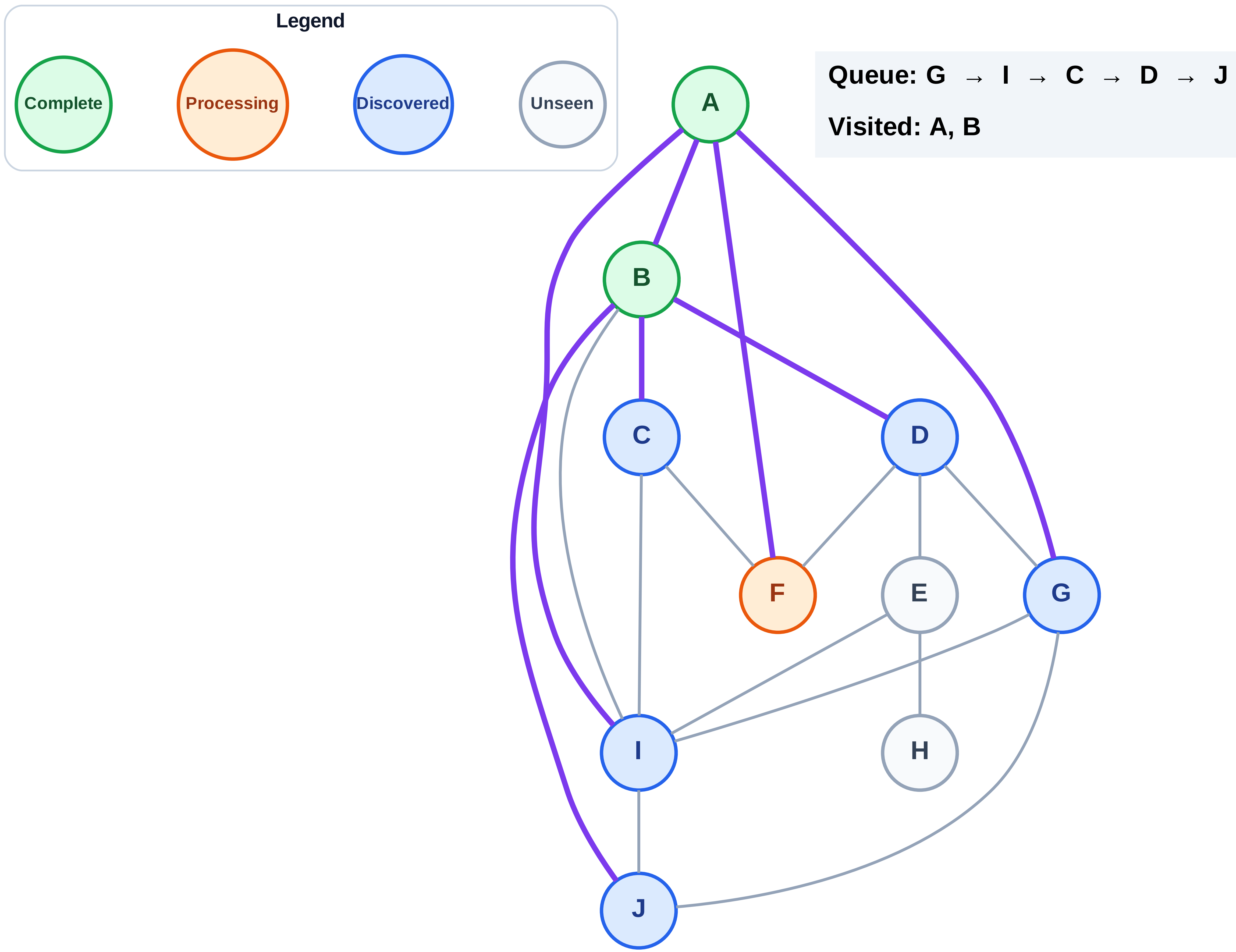
Queue: F → G → I → C → D → J

Visited: A, B



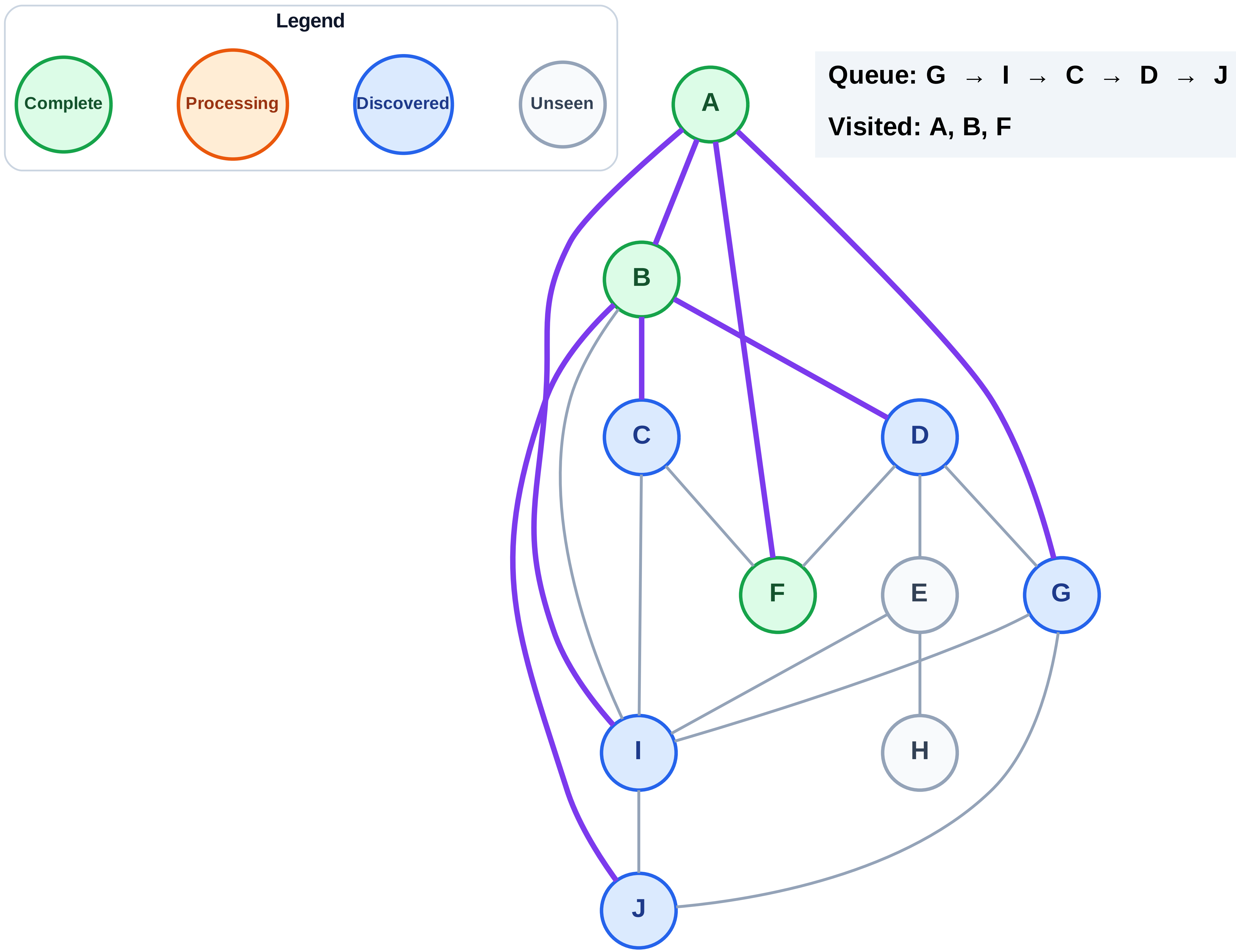
BFS Traversal

Step 6: Process node F



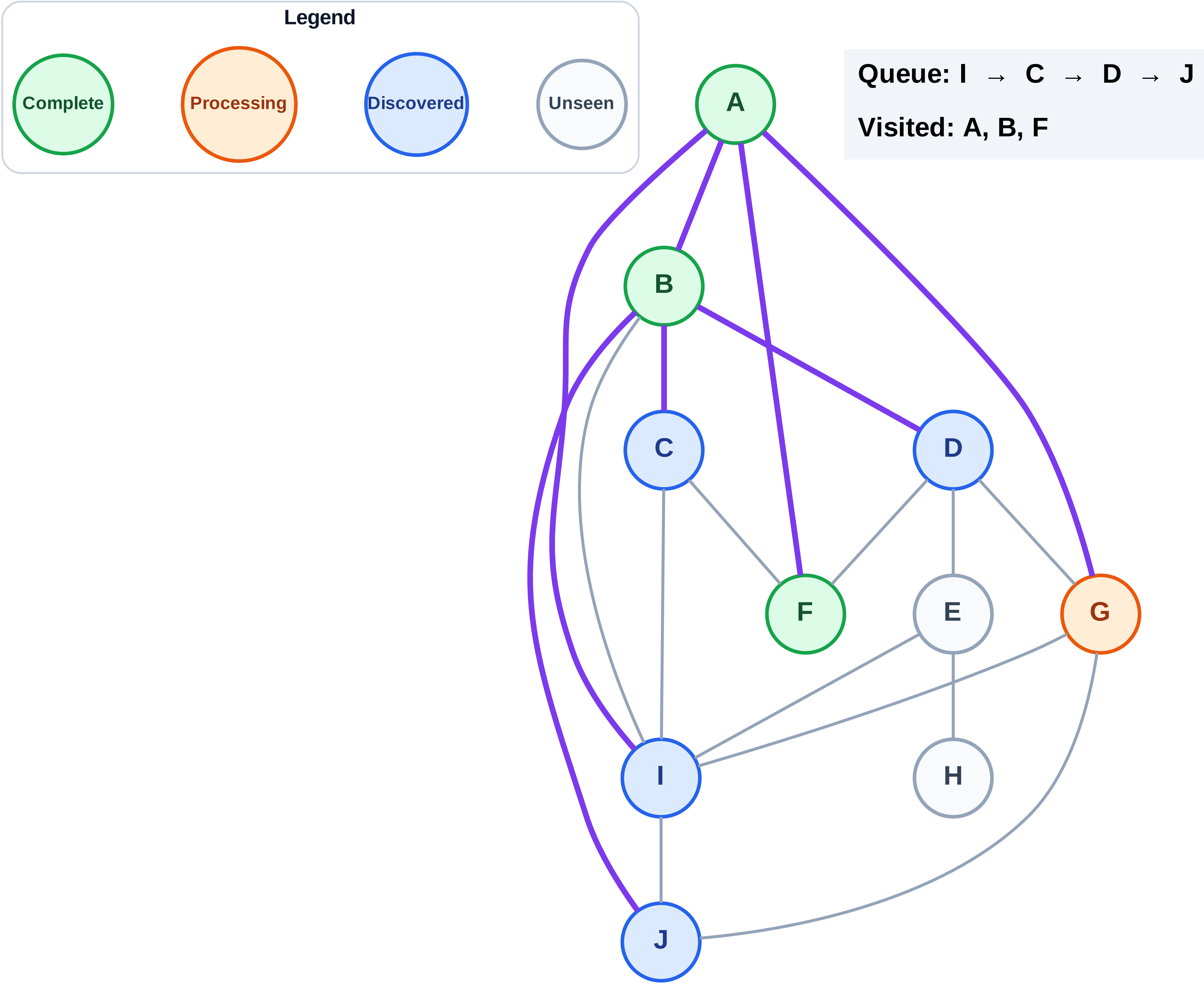
BFS Traversal

Step 7: No new neighbors from F



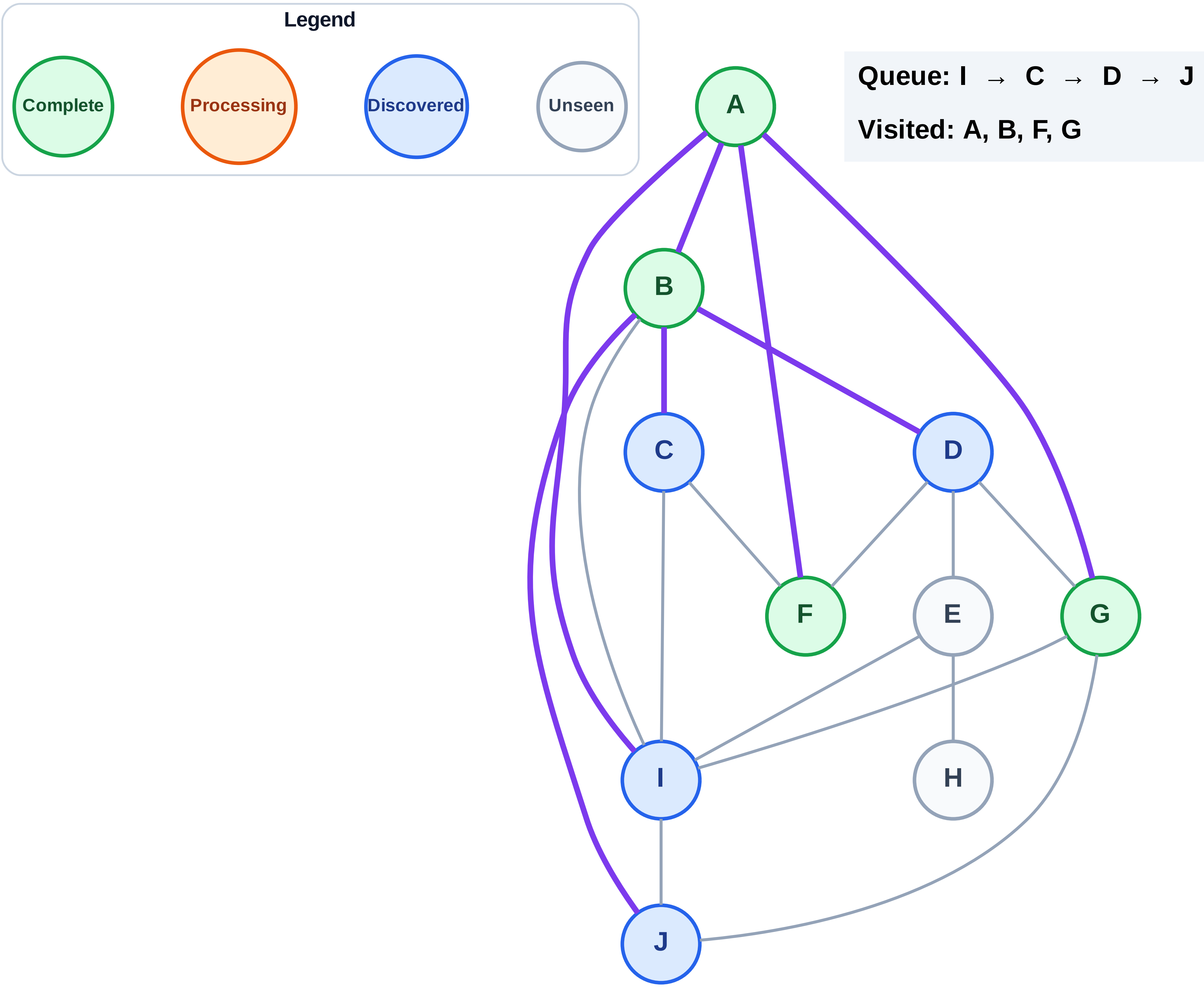
BFS Traversal

Step 8: Process node G



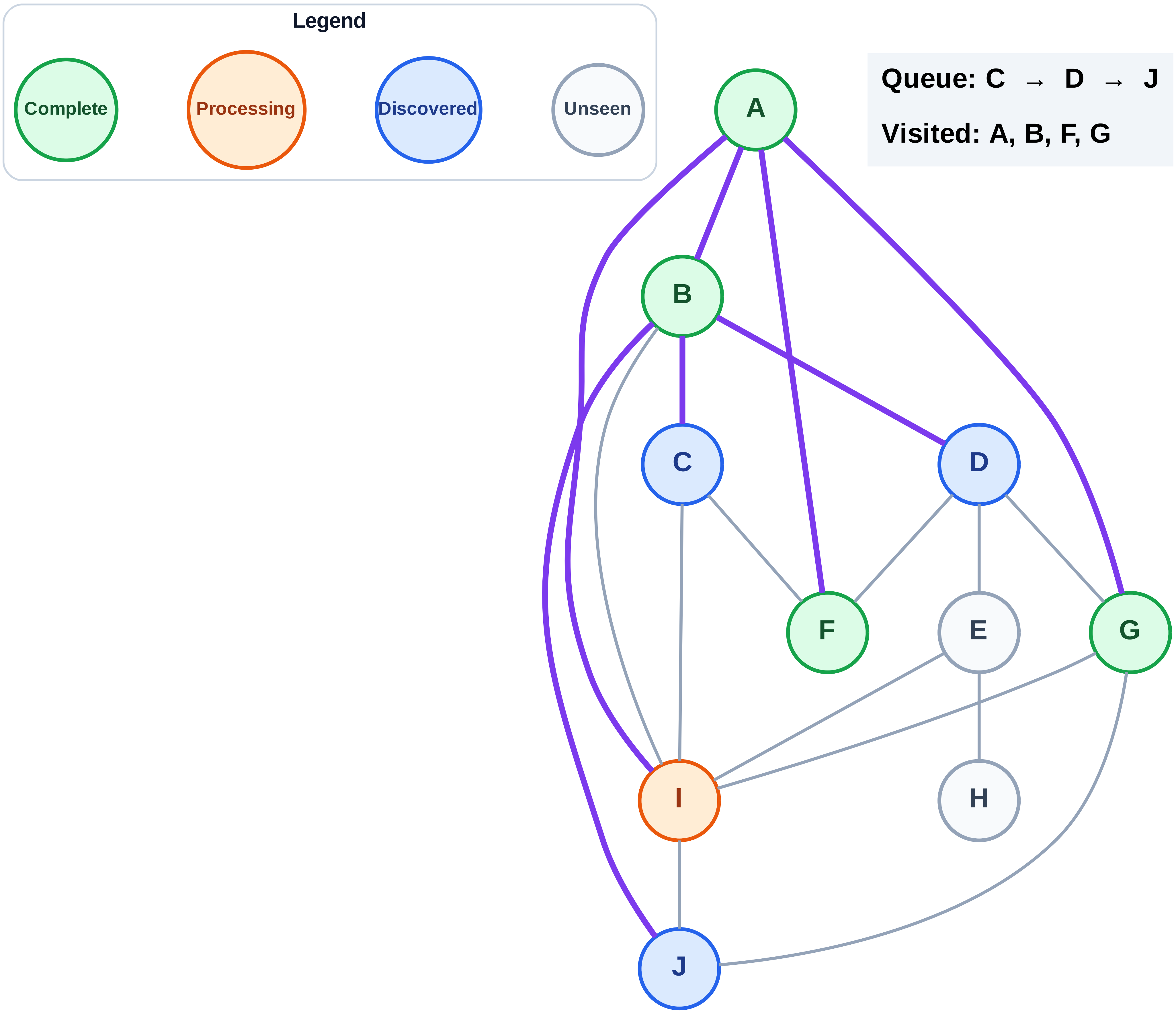
BFS Traversal

Step 9: No new neighbors from G



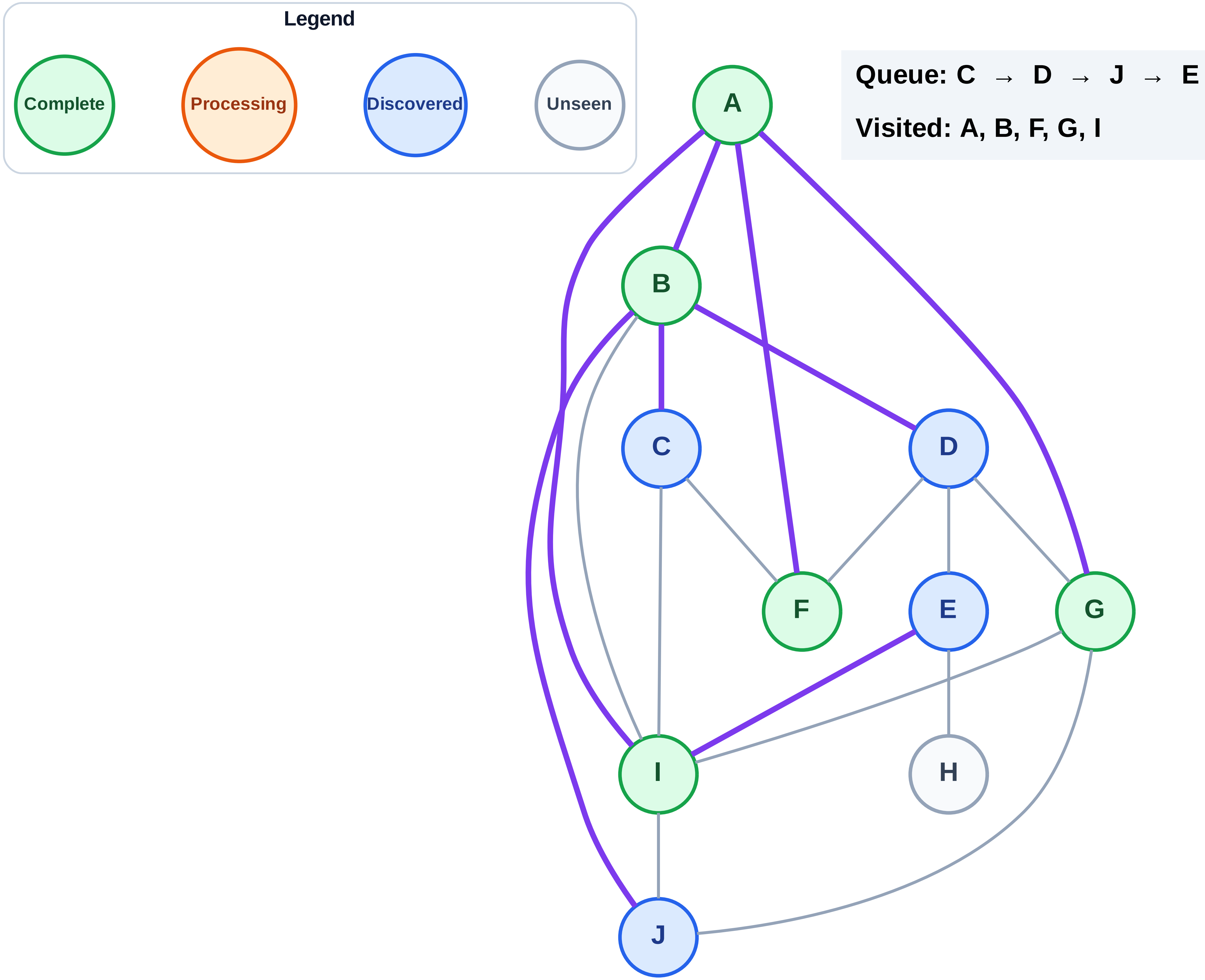
BFS Traversal

Step 10: Process node I



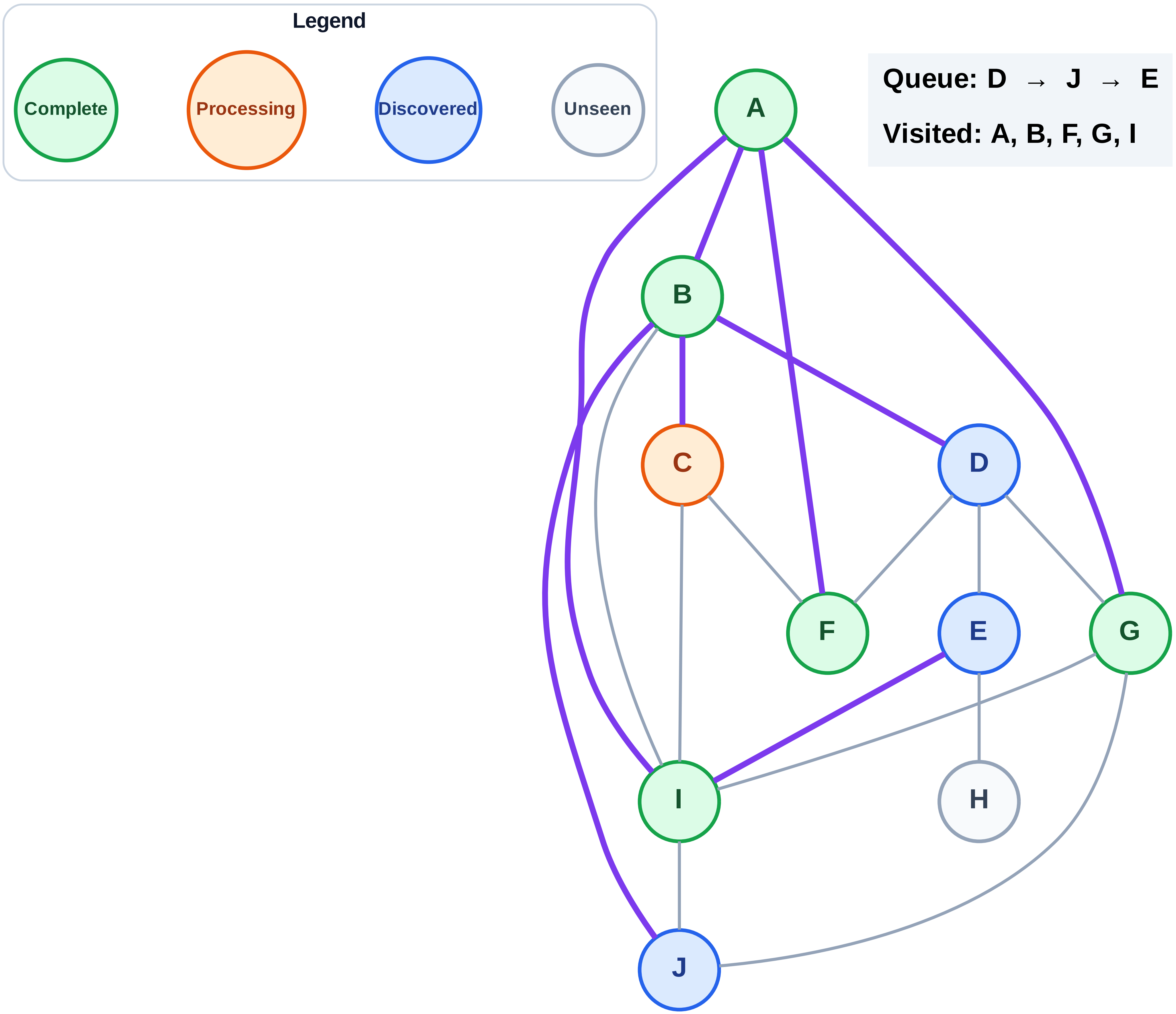
BFS Traversal

Step 11: Discover E from I



BFS Traversal

Step 12: Process node C

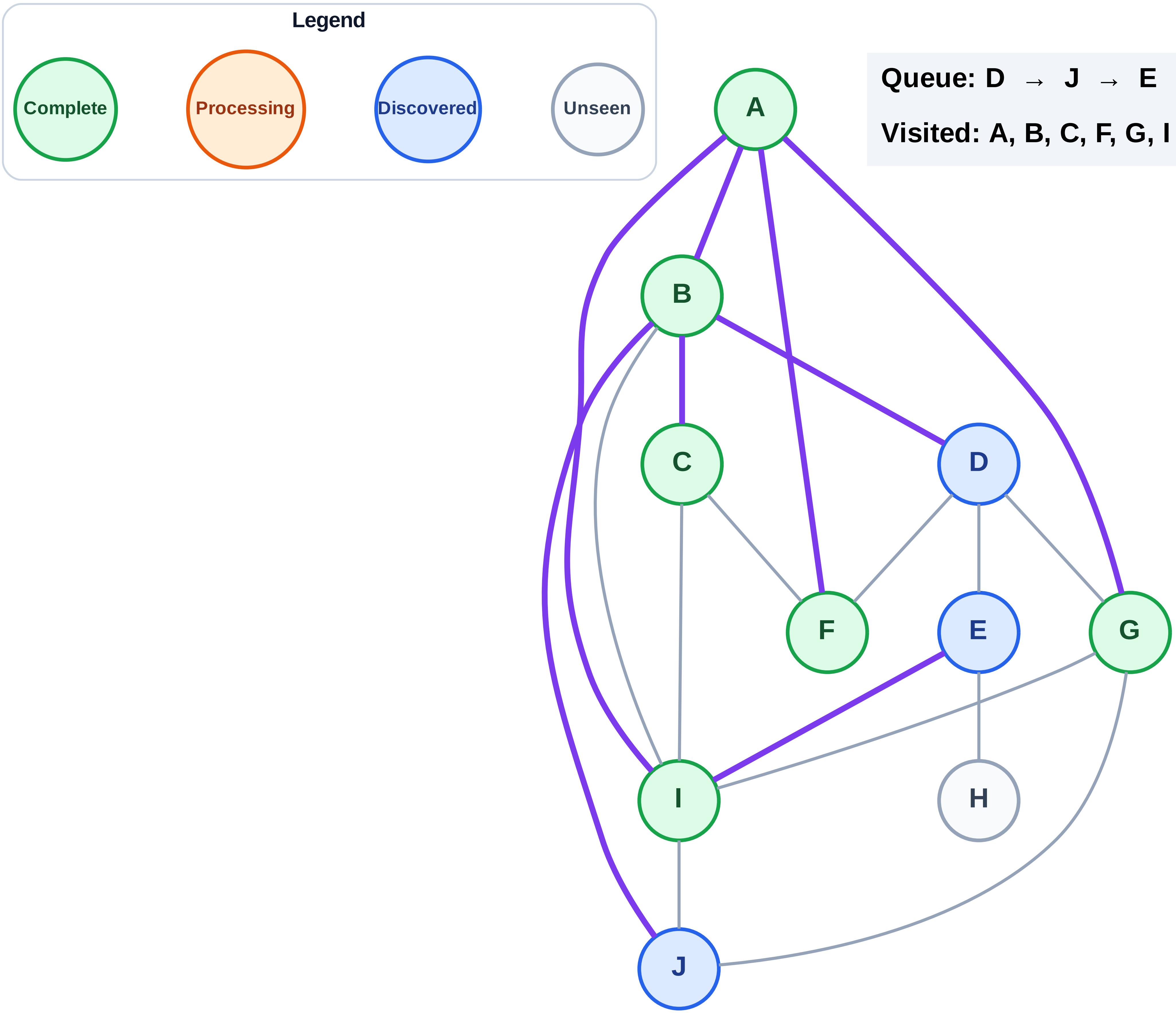


Queue: D → J → E

Visited: A, B, F, G, I

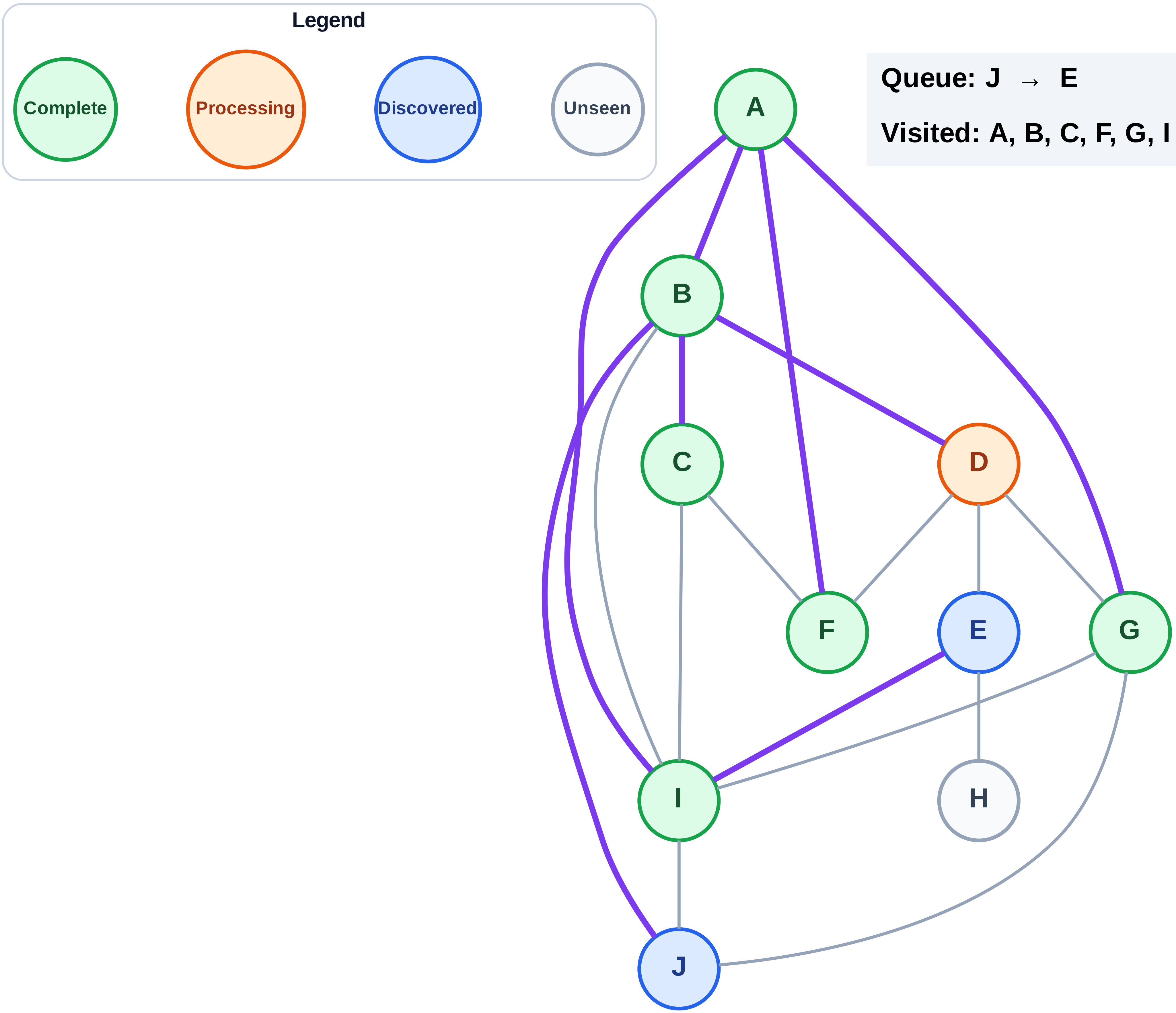
BFS Traversal

Step 13: No new neighbors from C



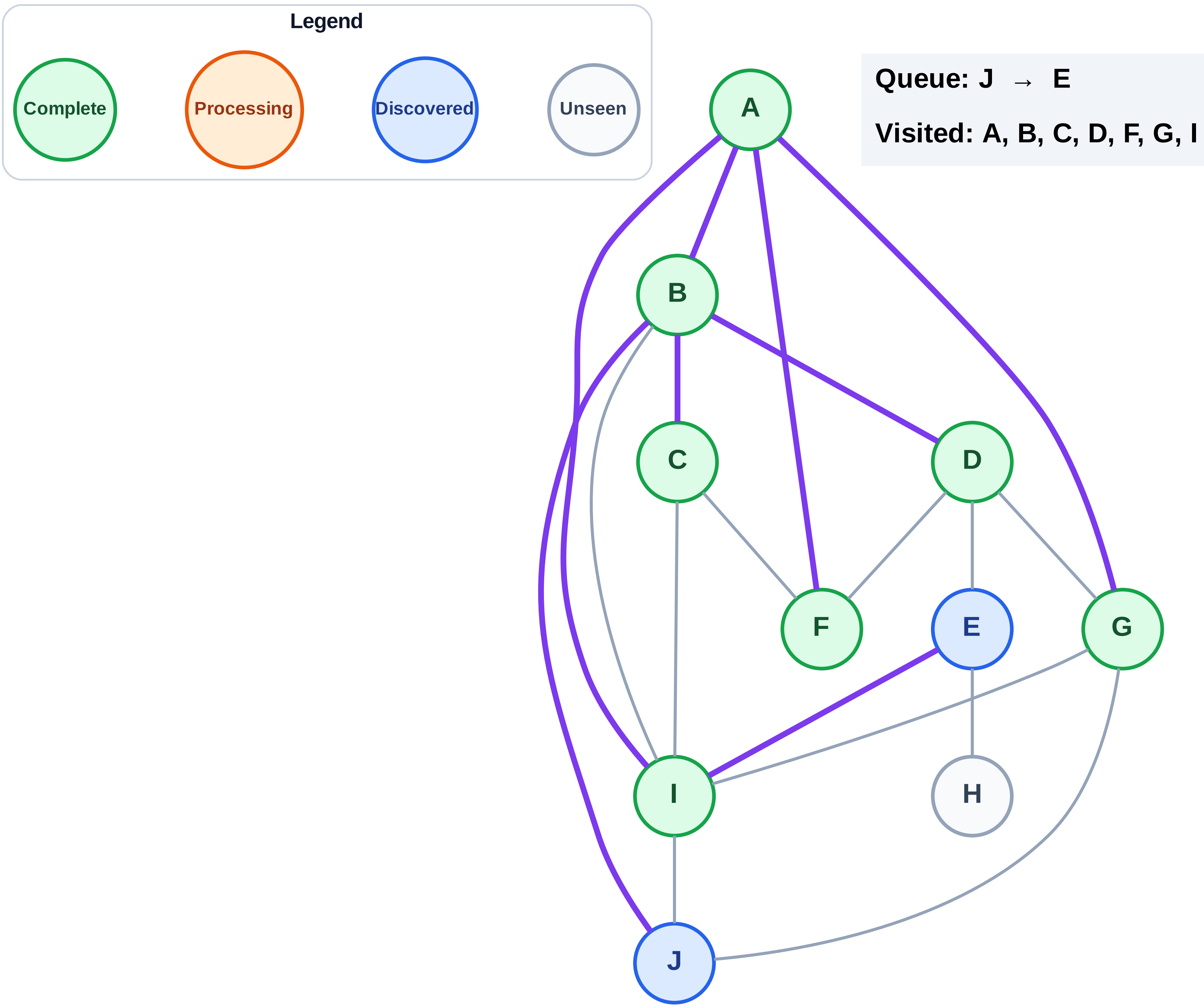
BFS Traversal

Step 14: Process node D



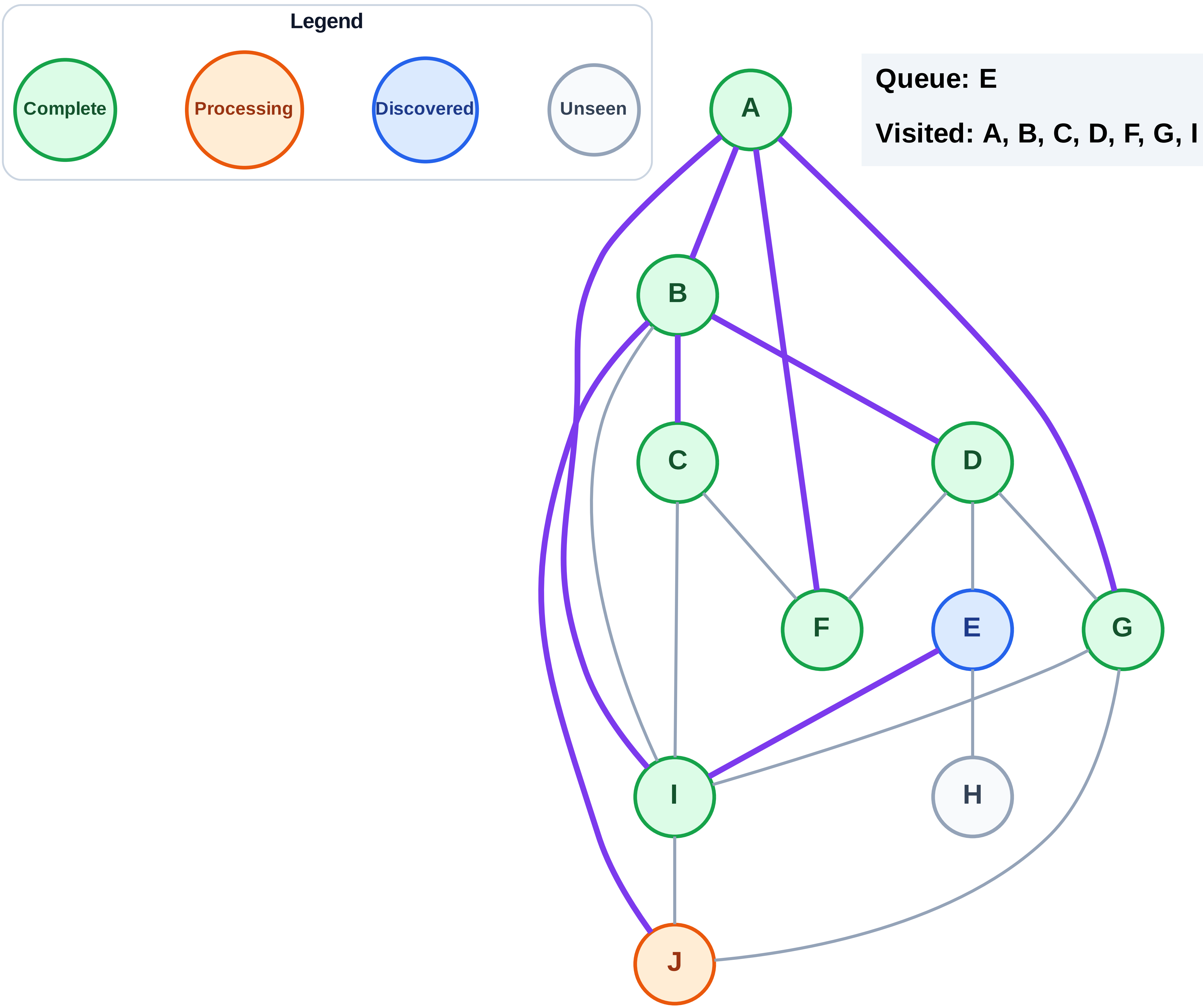
BFS Traversal

Step 15: No new neighbors from D



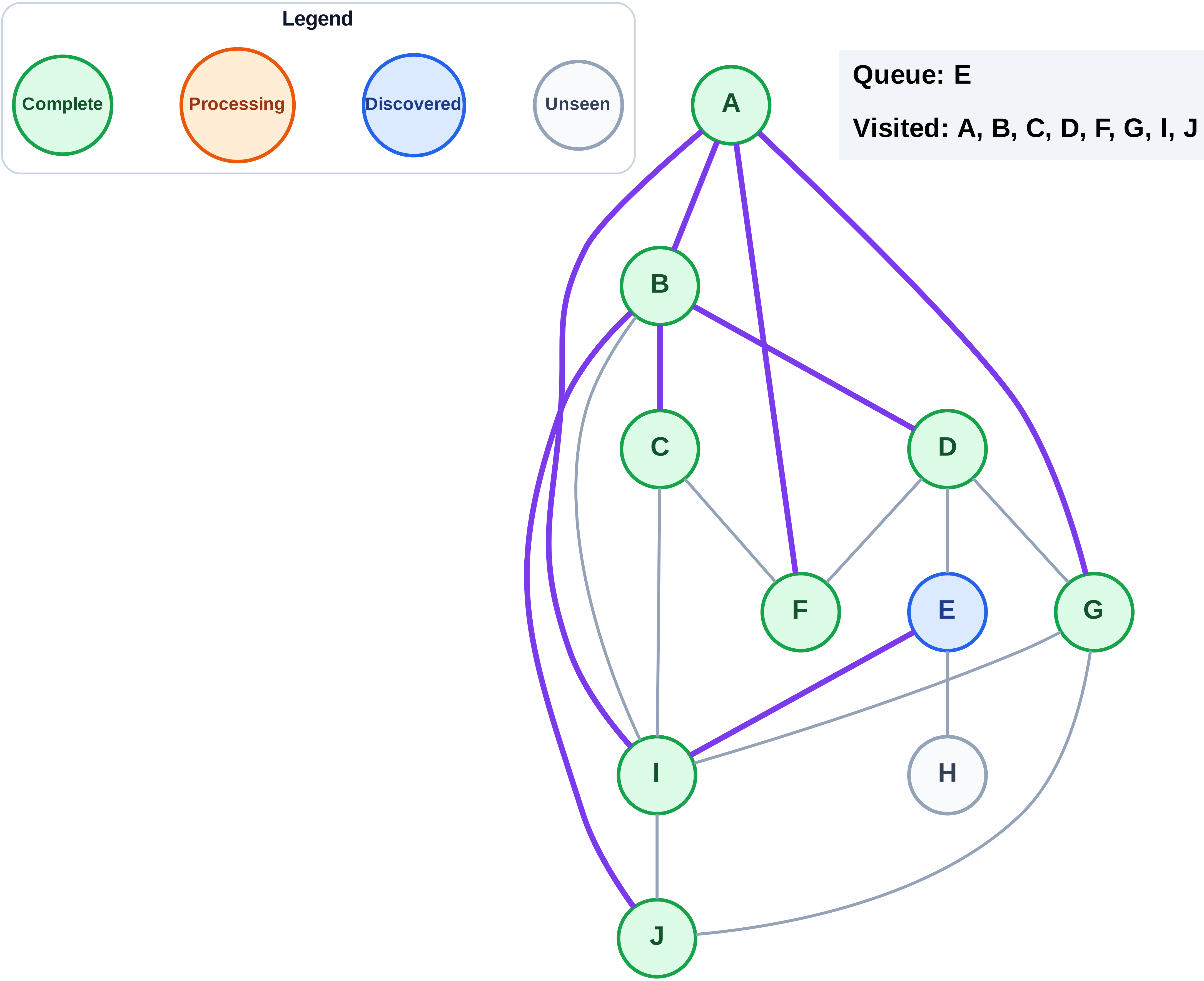
BFS Traversal

Step 16: Process node J



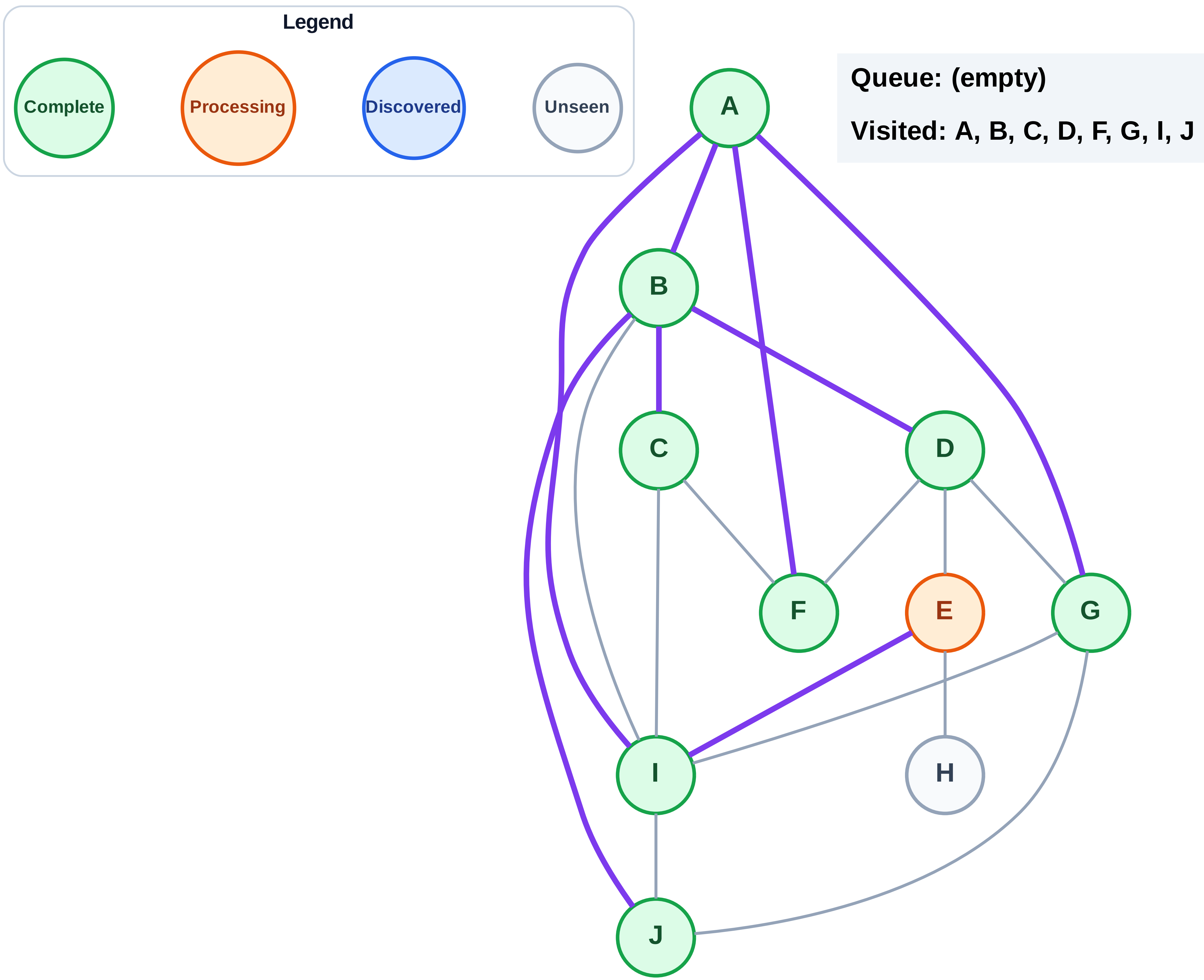
BFS Traversal

Step 17: No new neighbors from J



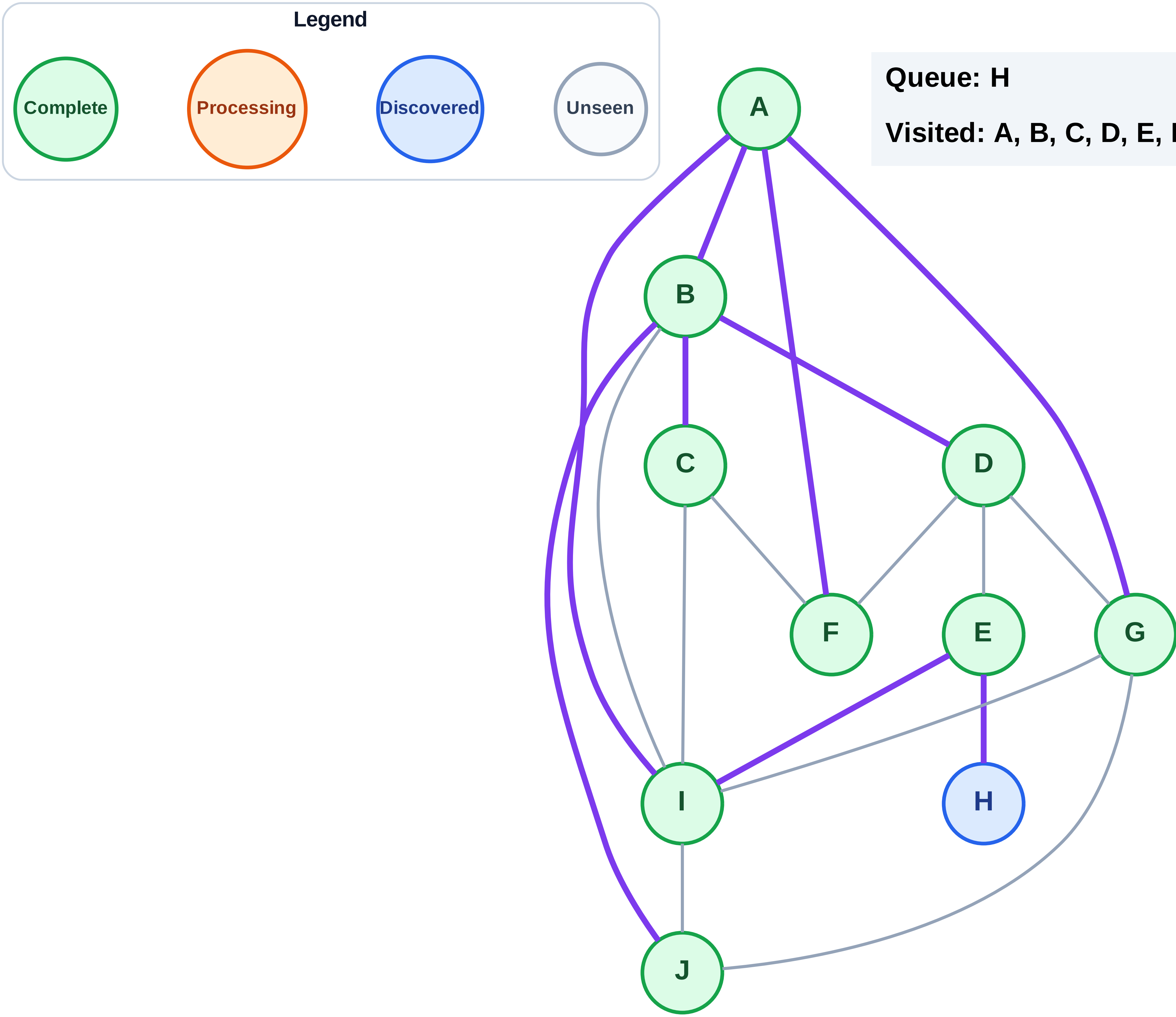
BFS Traversal

Step 18: Process node E



BFS Traversal

Step 19: Discover H from E

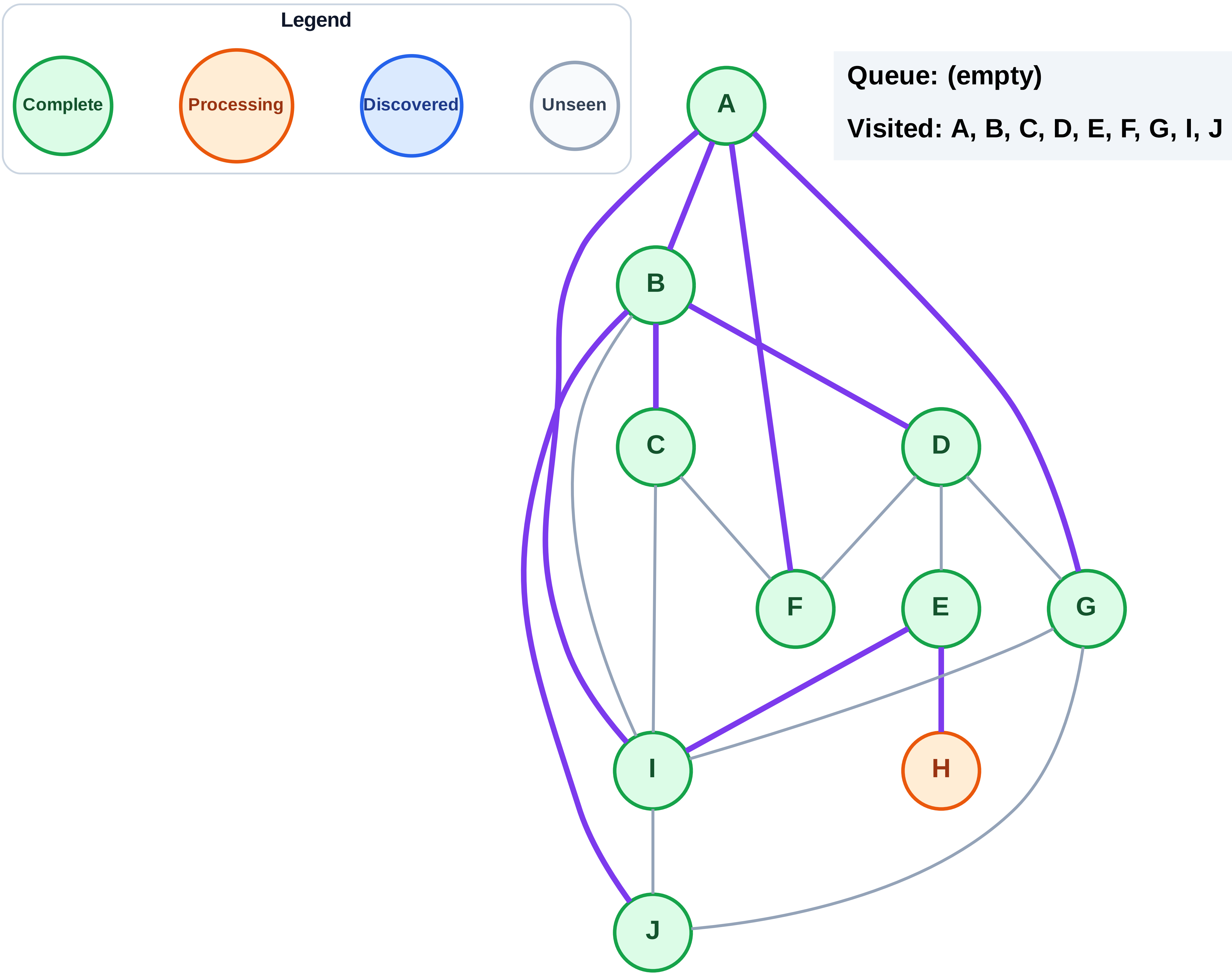


Queue: H

Visited: A, B, C, D, E, F, G, I, J

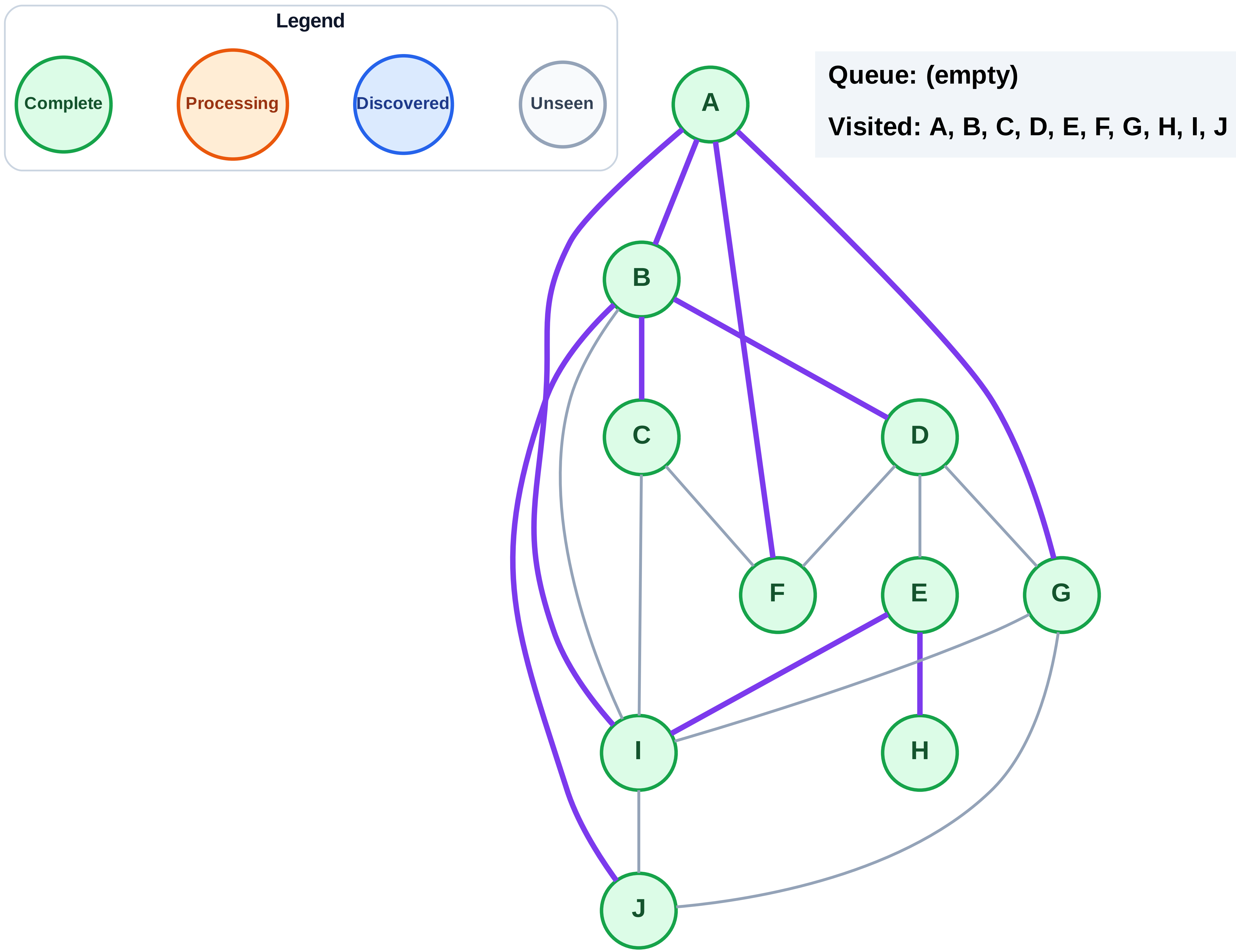
BFS Traversal

Step 20: Process node H



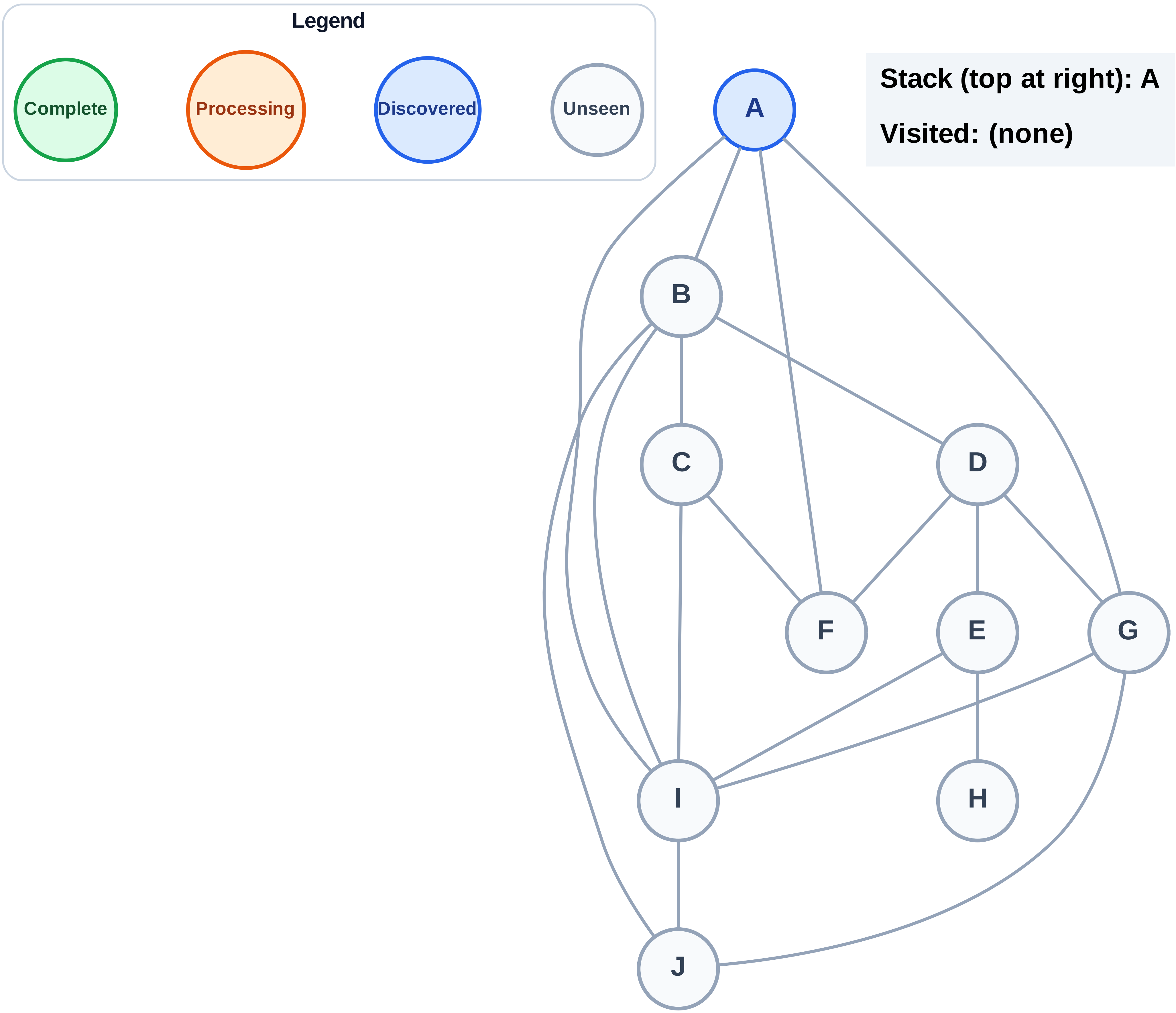
BFS Traversal

Step 21: No new neighbors from H



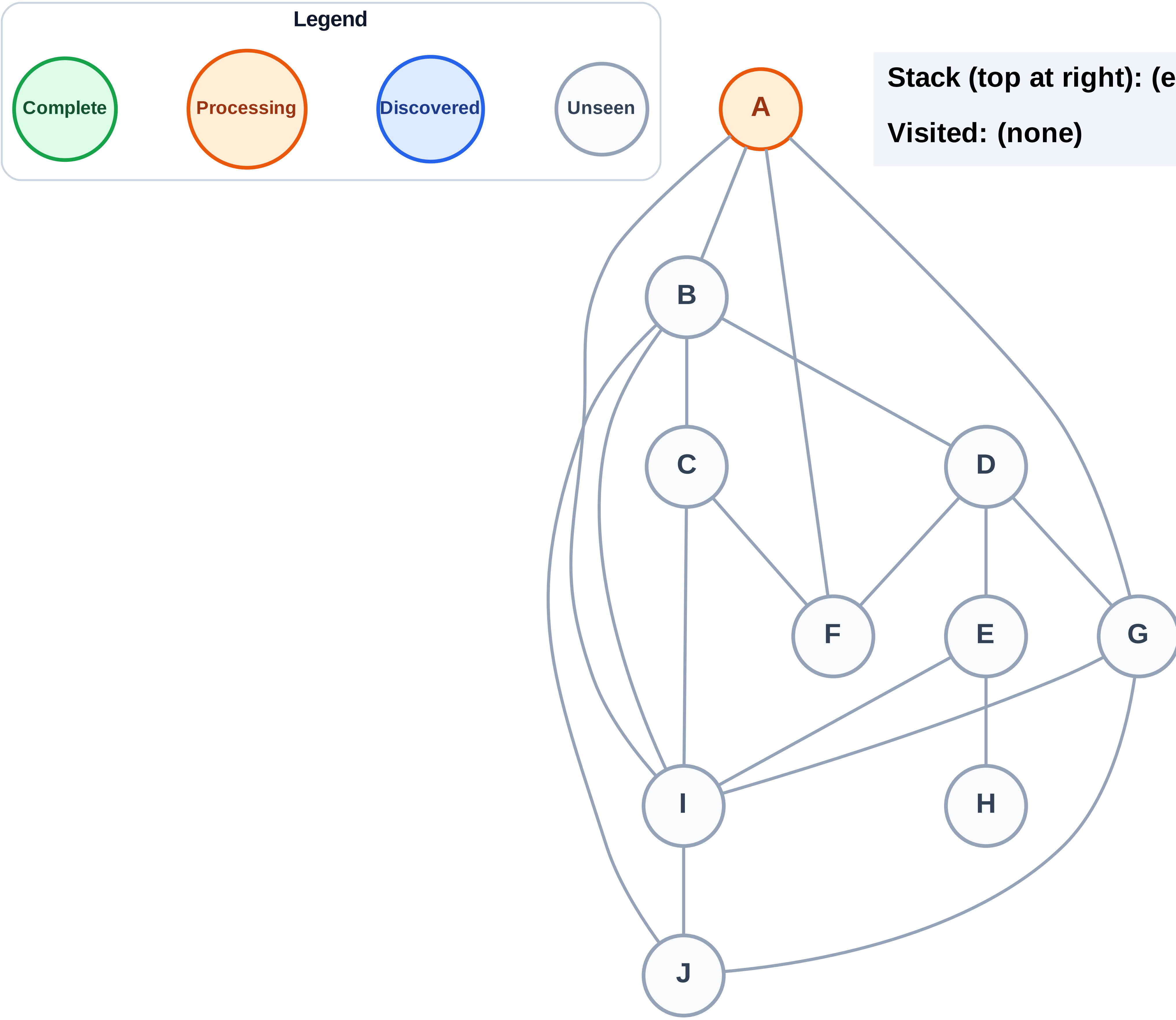
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

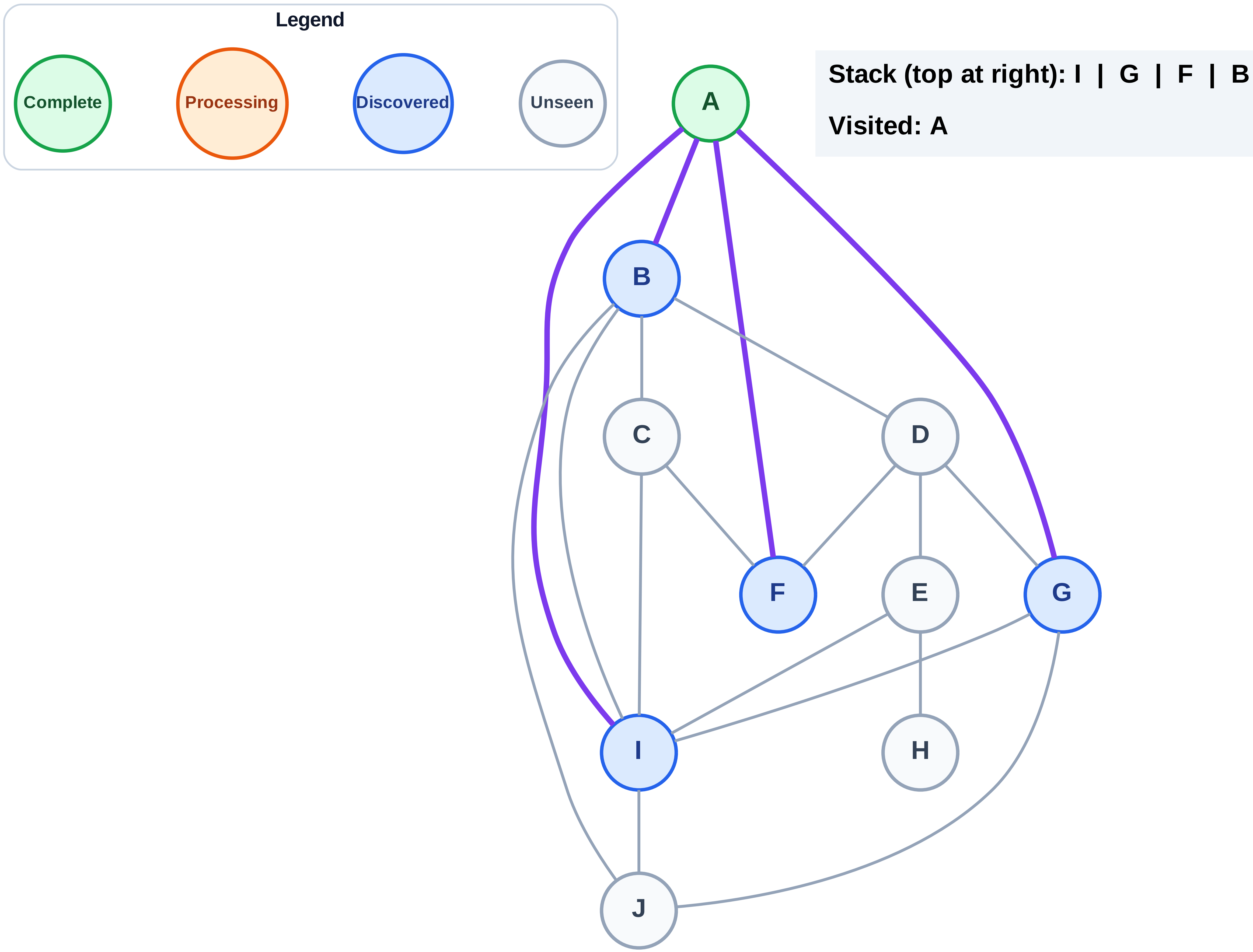


Stack (top at right): (empty)

Visited: (none)

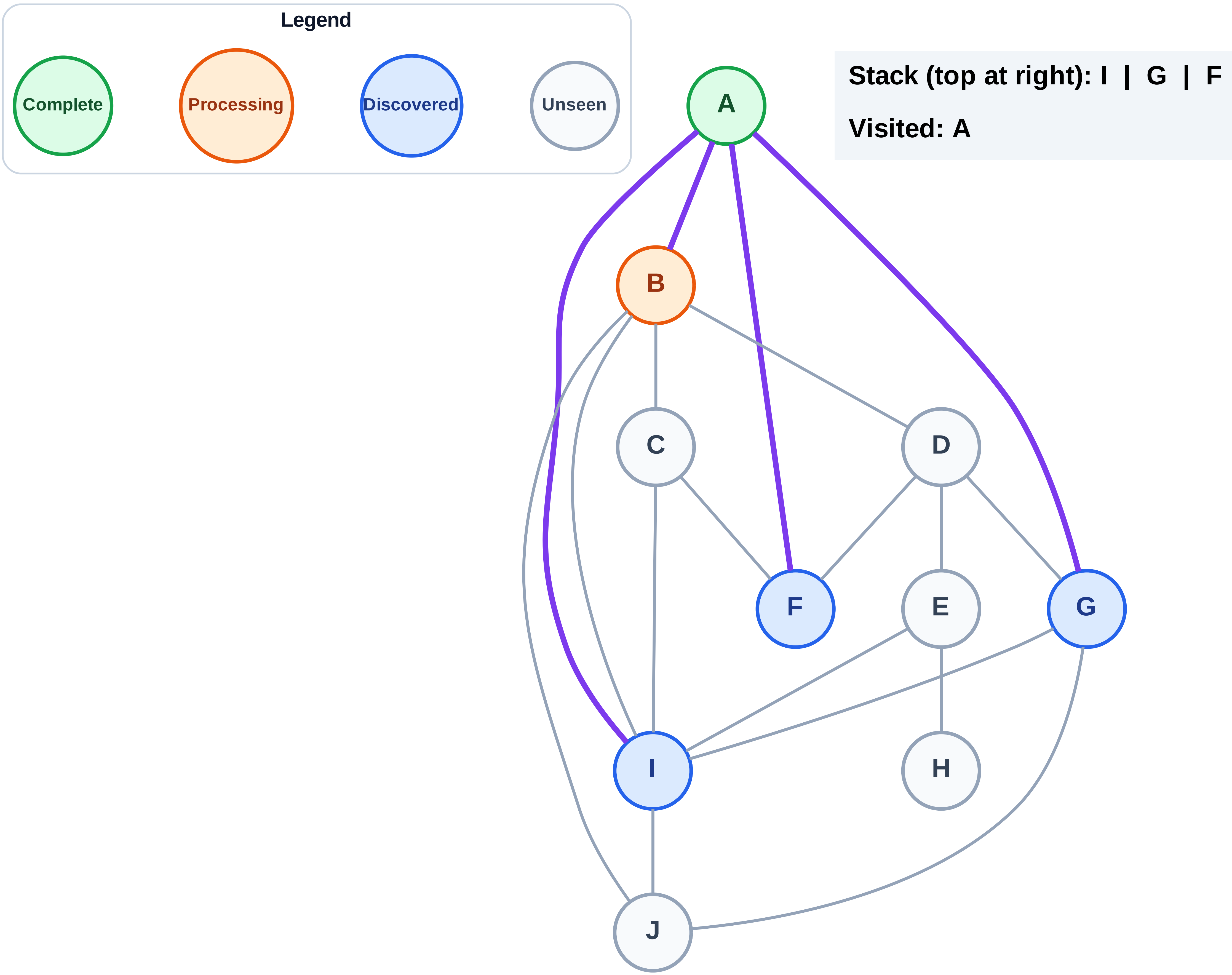
DFS Traversal

Step 3: Discover I, G, F, B from A



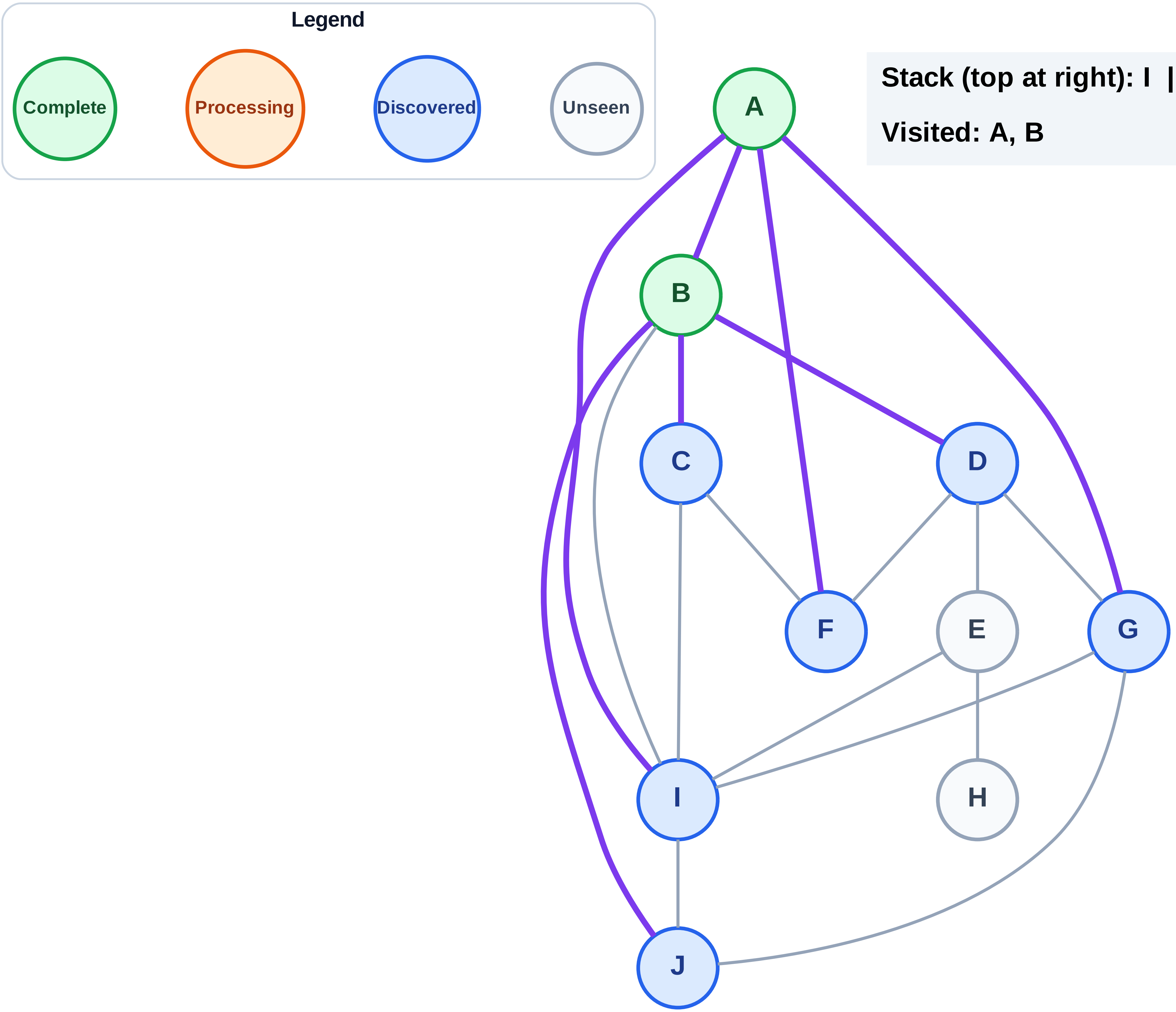
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover J, D, C from B

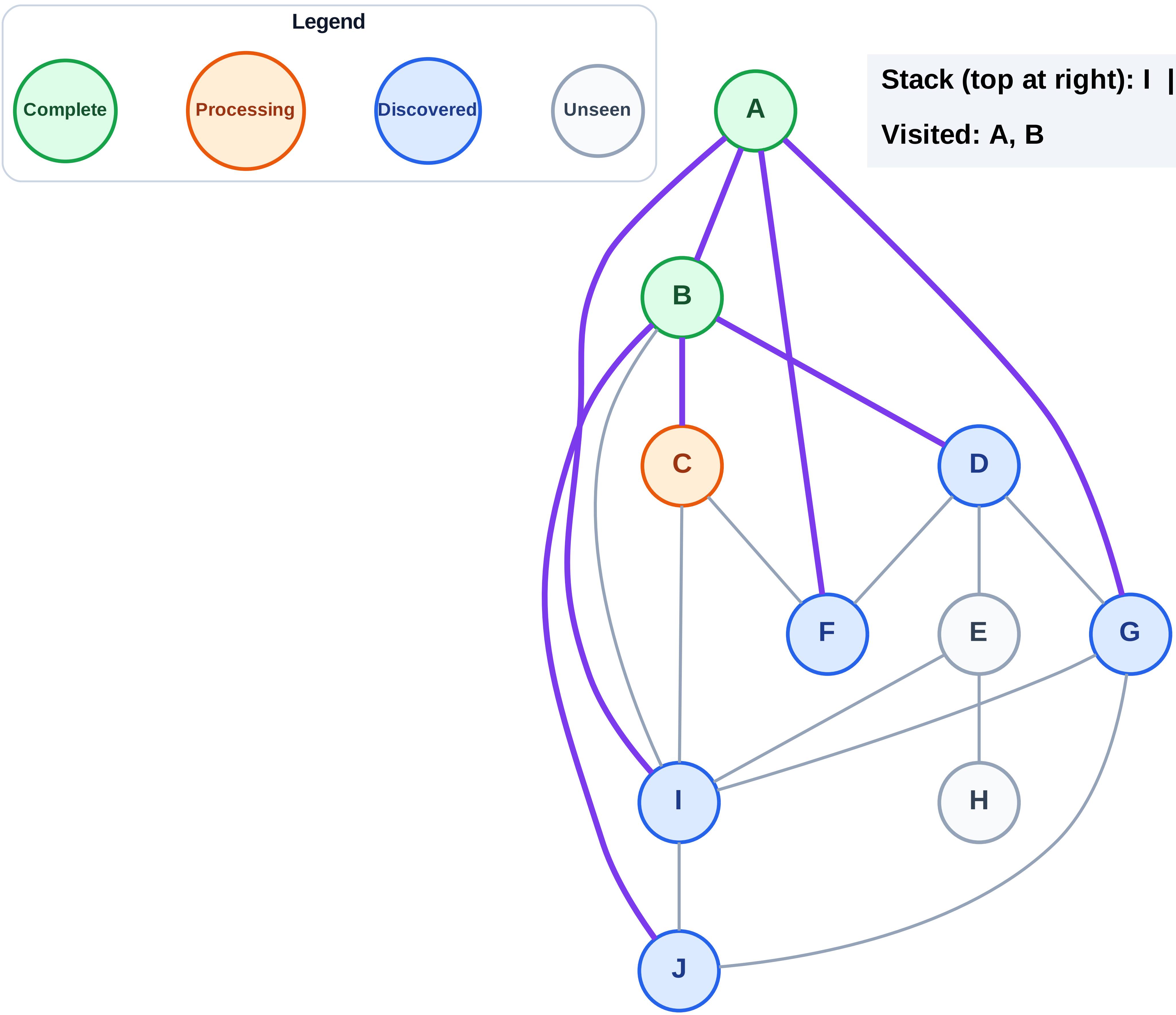


Stack (top at right): I | G | F | J | D | C

Visited: A, B

DFS Traversal

Step 6: Process node C

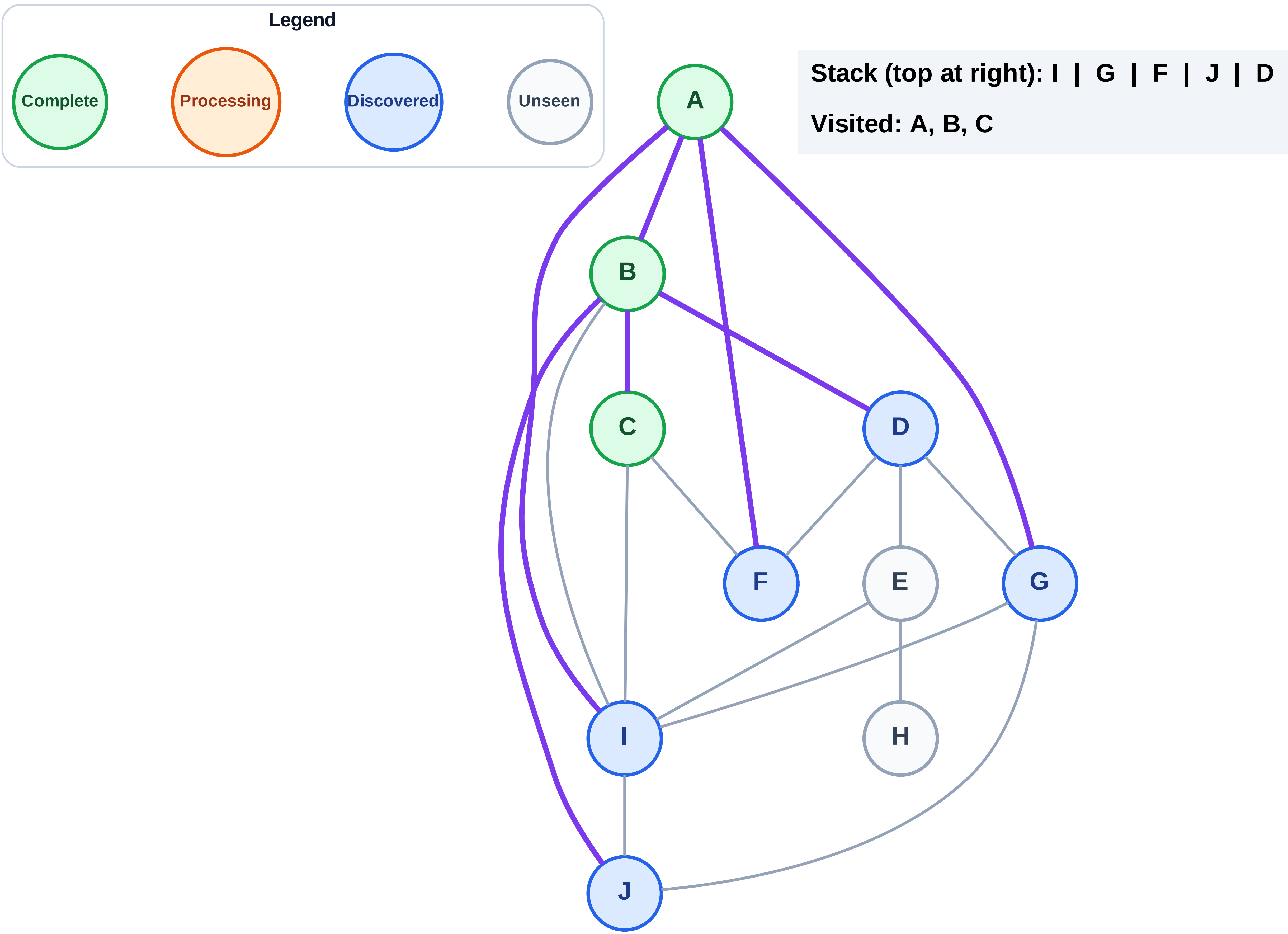


Stack (top at right): I | G | F | J | D

Visited: A, B

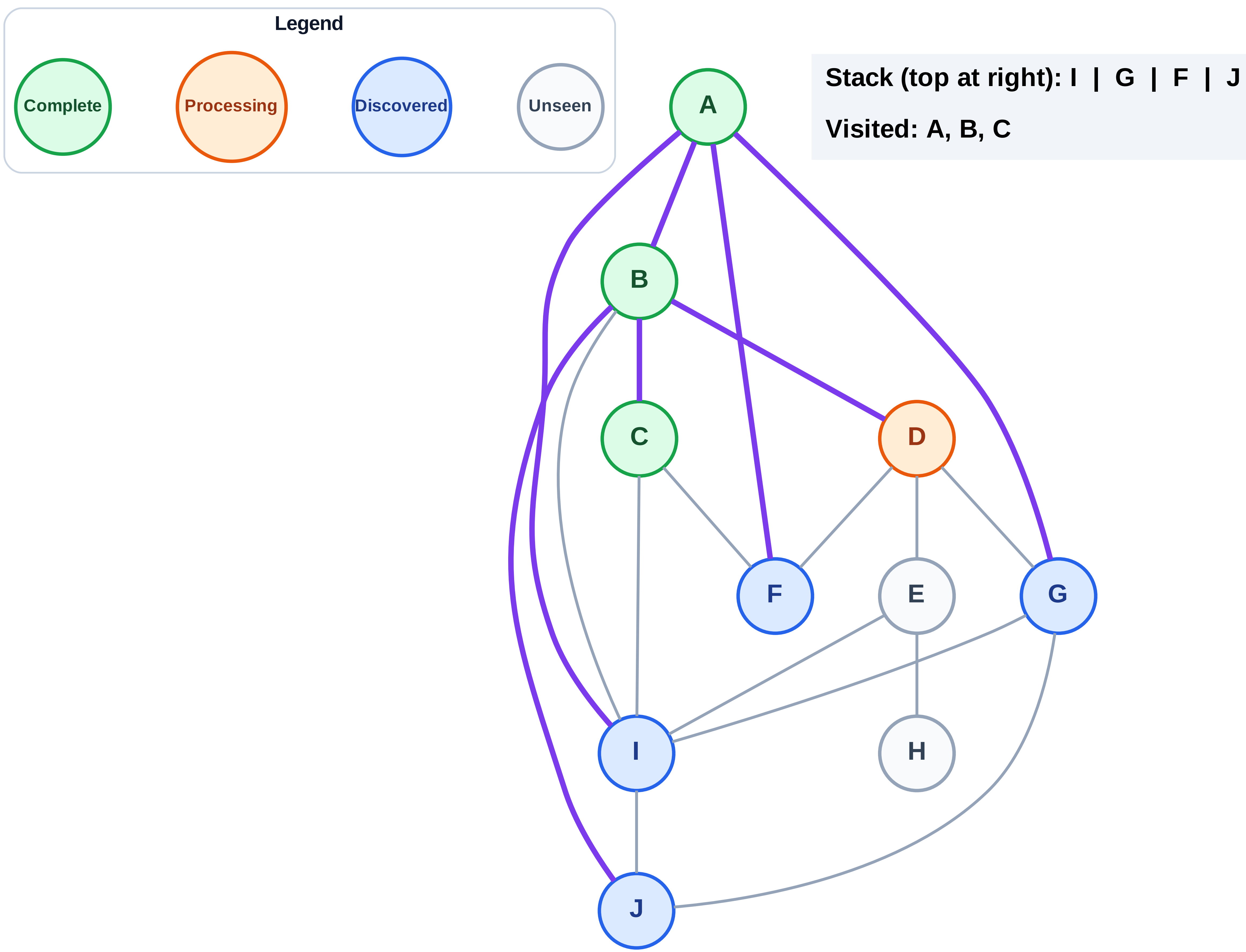
DFS Traversal

Step 7: No new neighbors from C



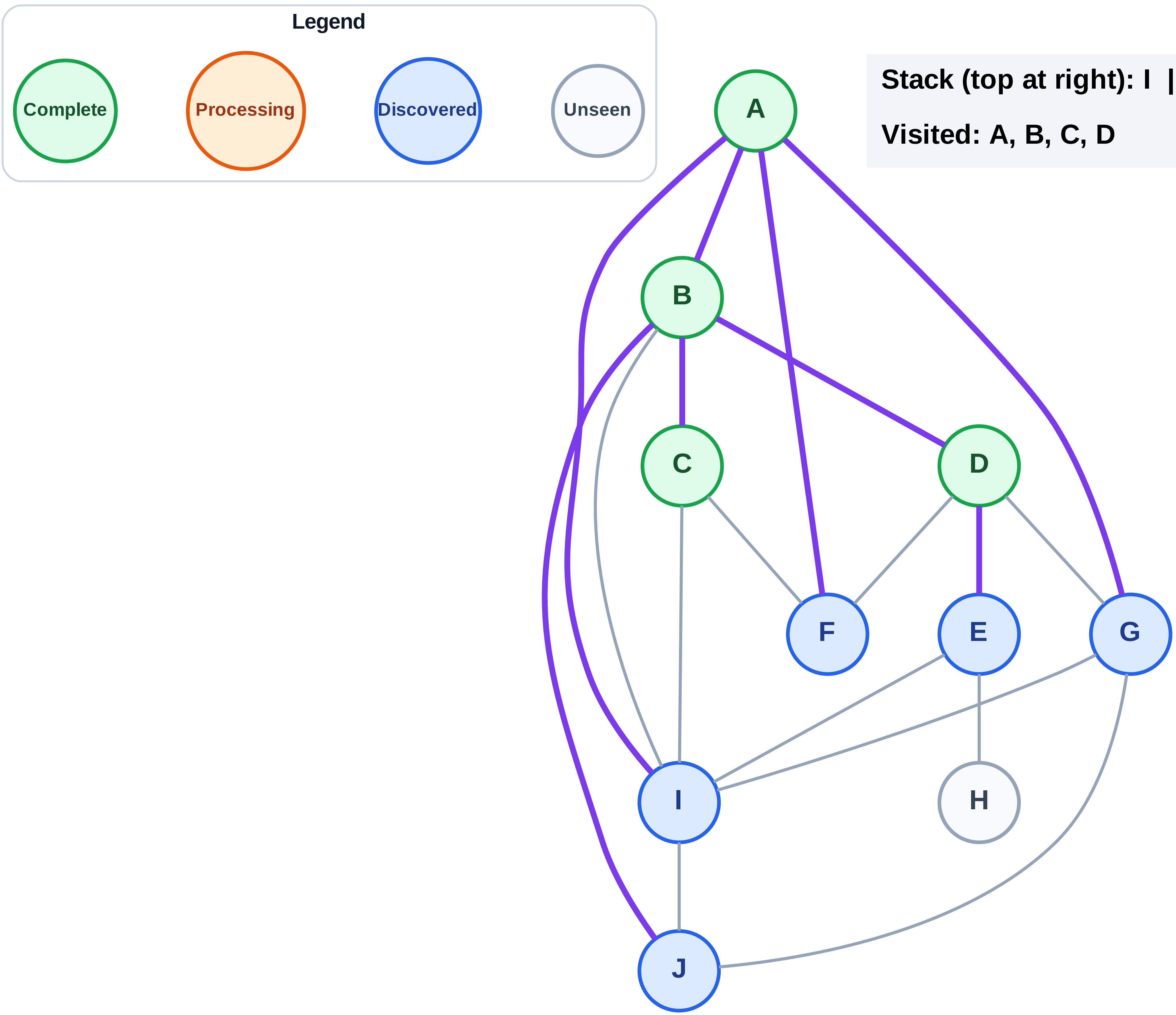
DFS Traversal

Step 8: Process node D



DFS Traversal

Step 9: Discover E from D

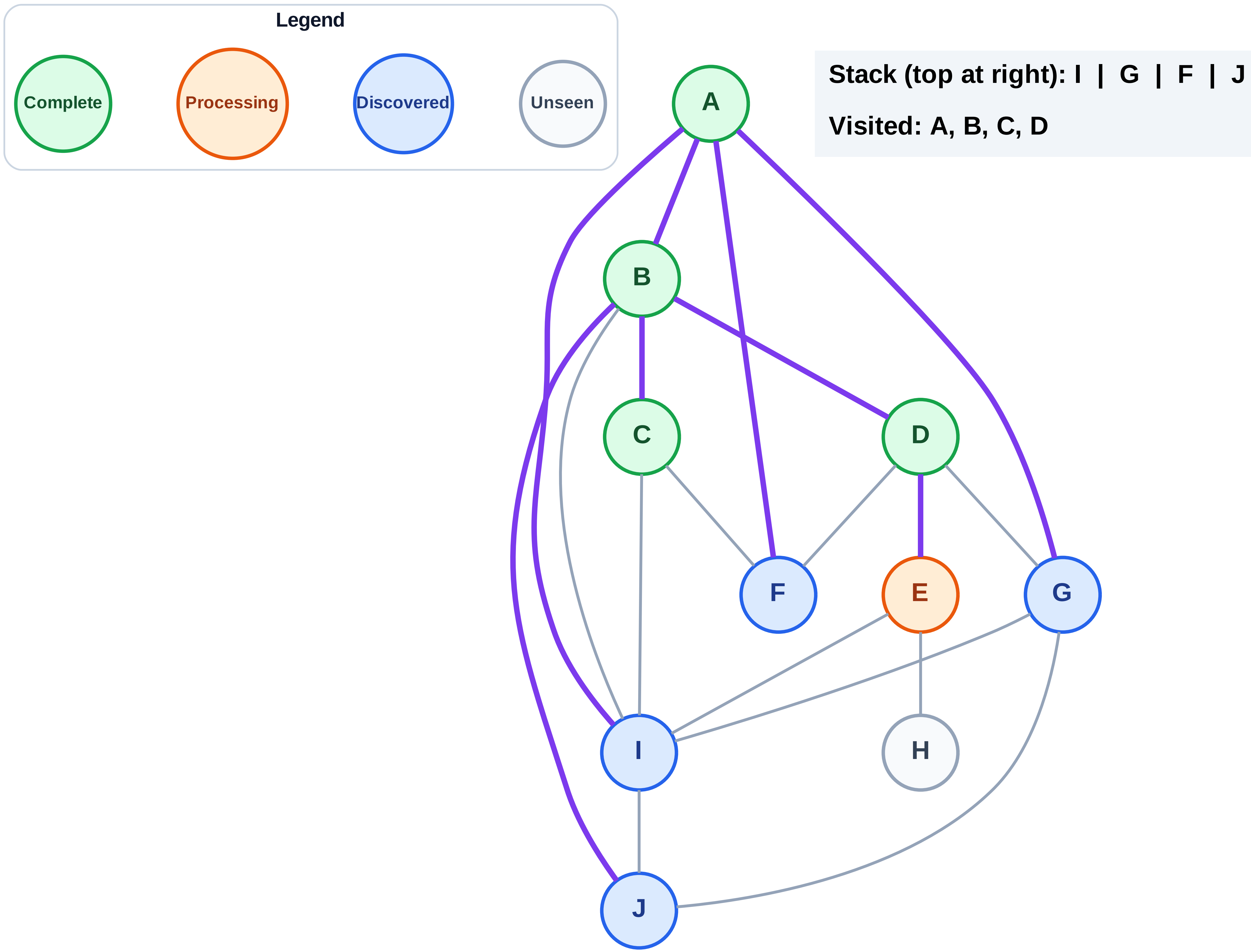


Stack (top at right): I | G | F | J | E

Visited: A, B, C, D

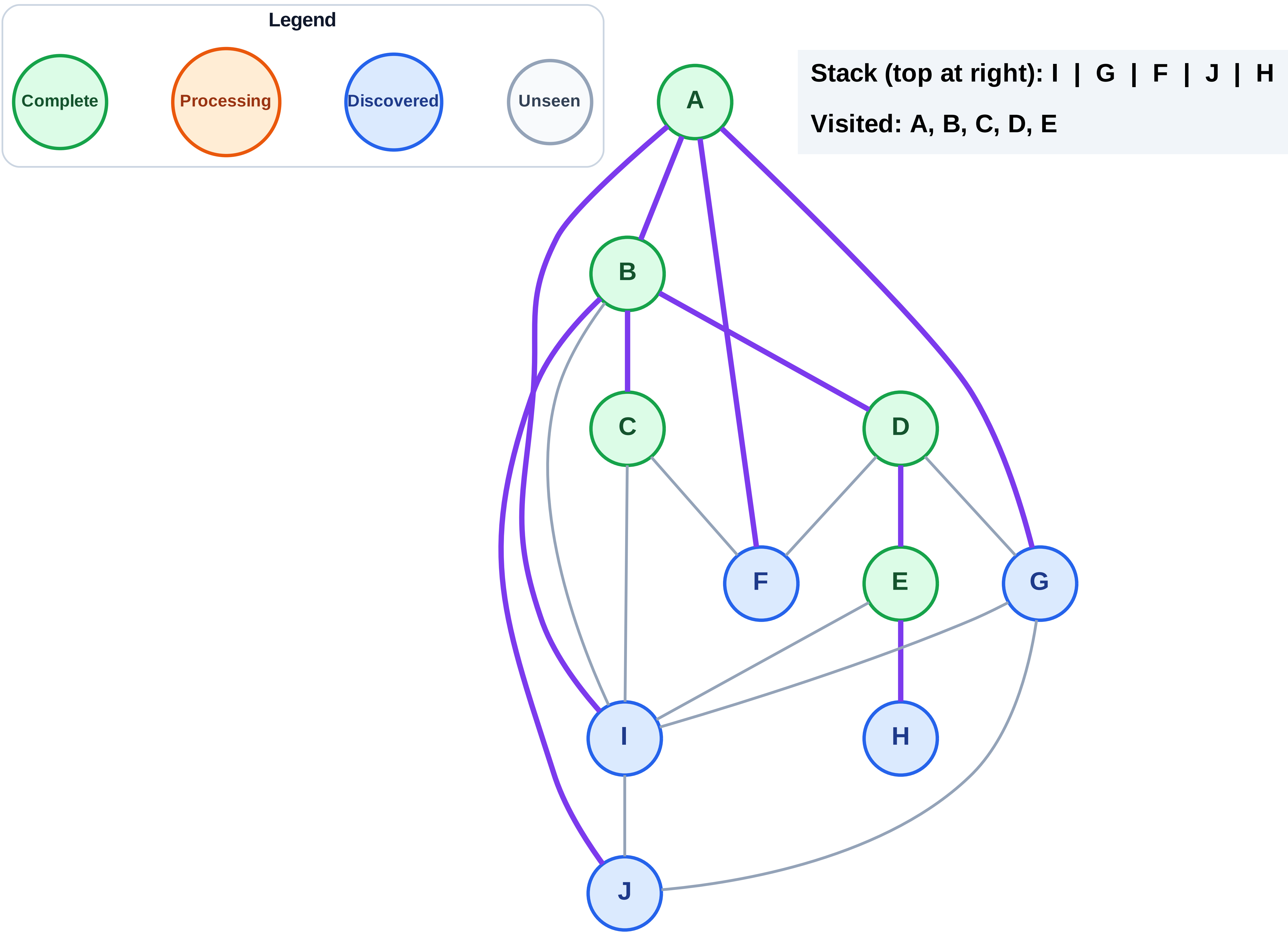
DFS Traversal

Step 10: Process node E



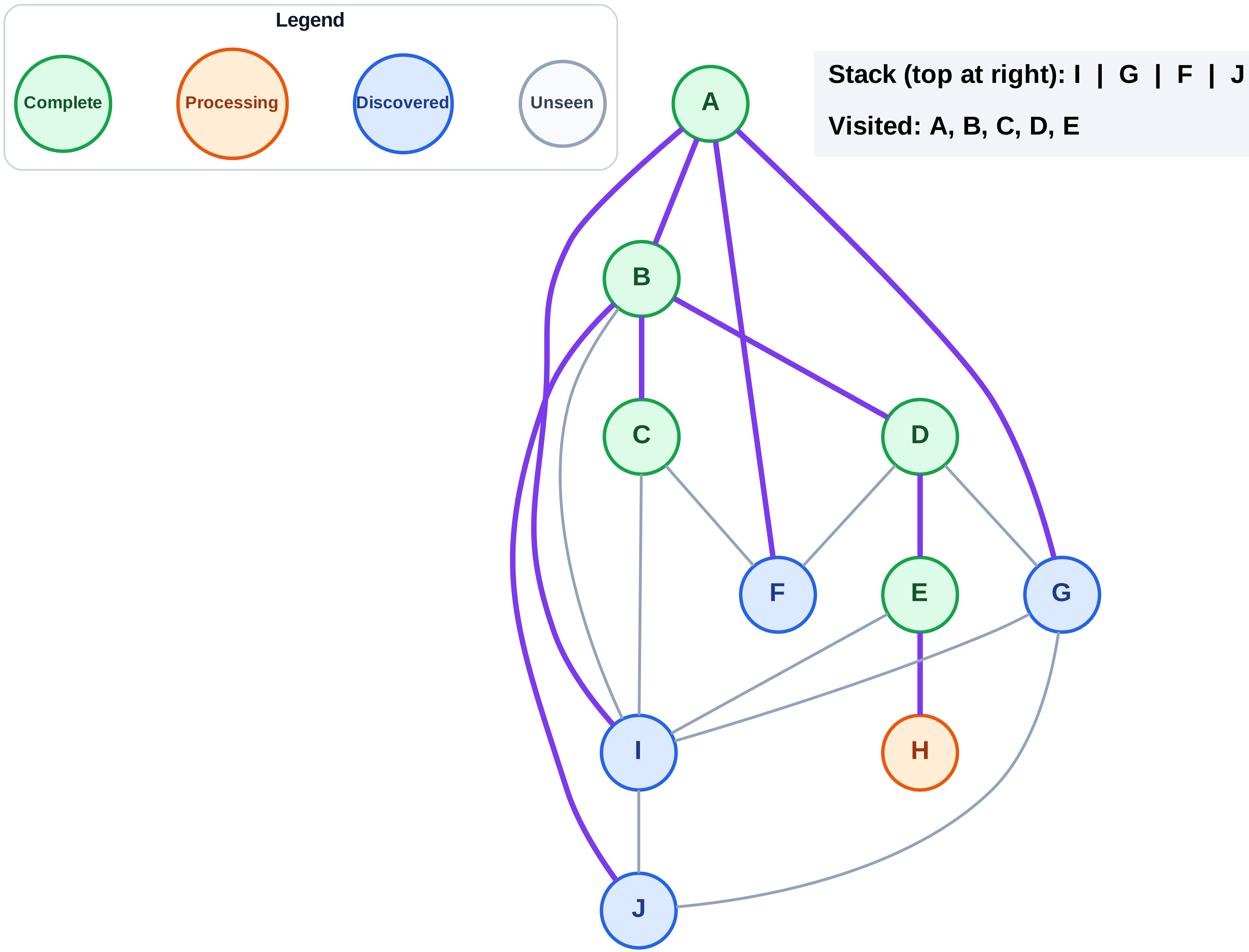
DFS Traversal

Step 11: Discover H from E



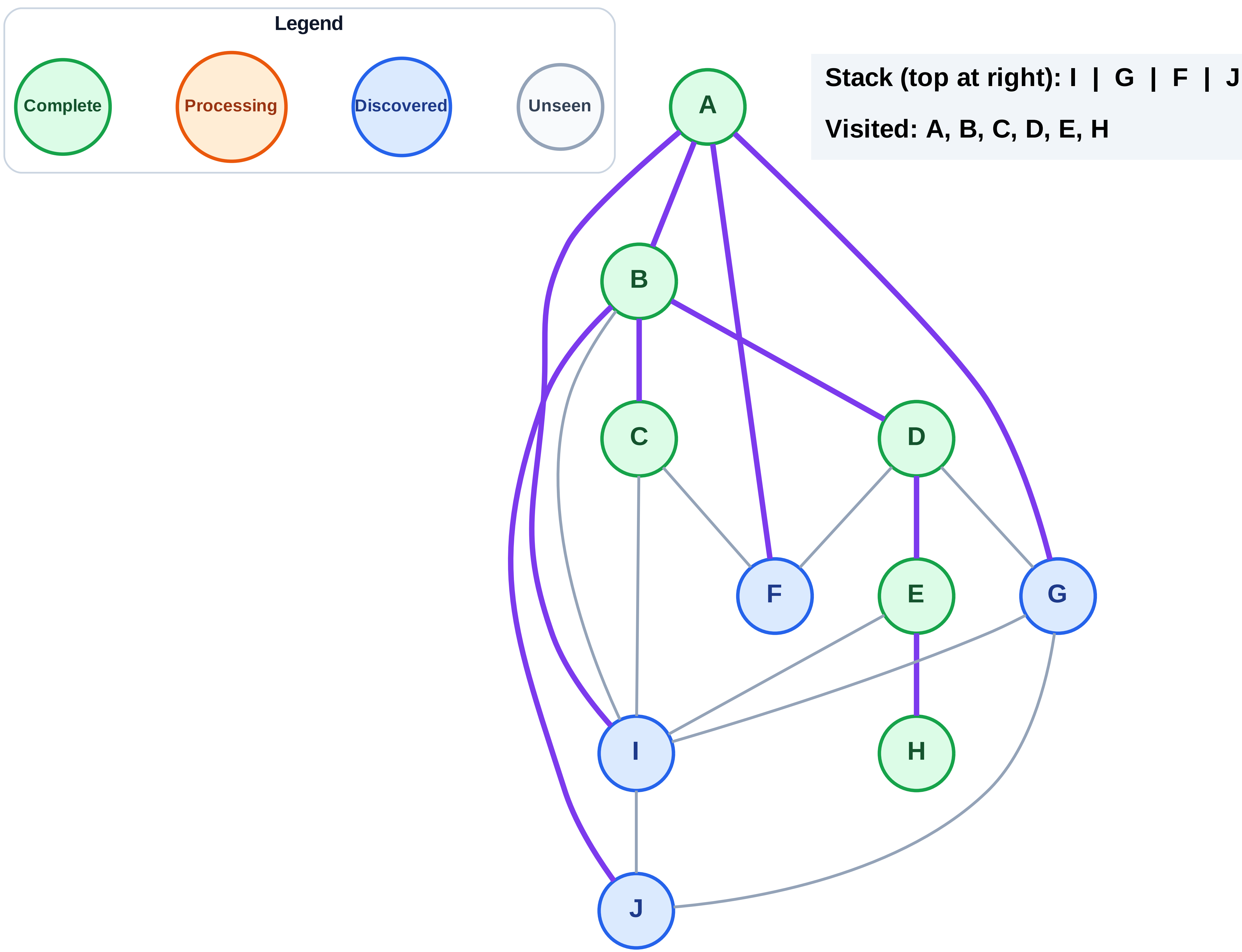
DFS Traversal

Step 12: Process node H



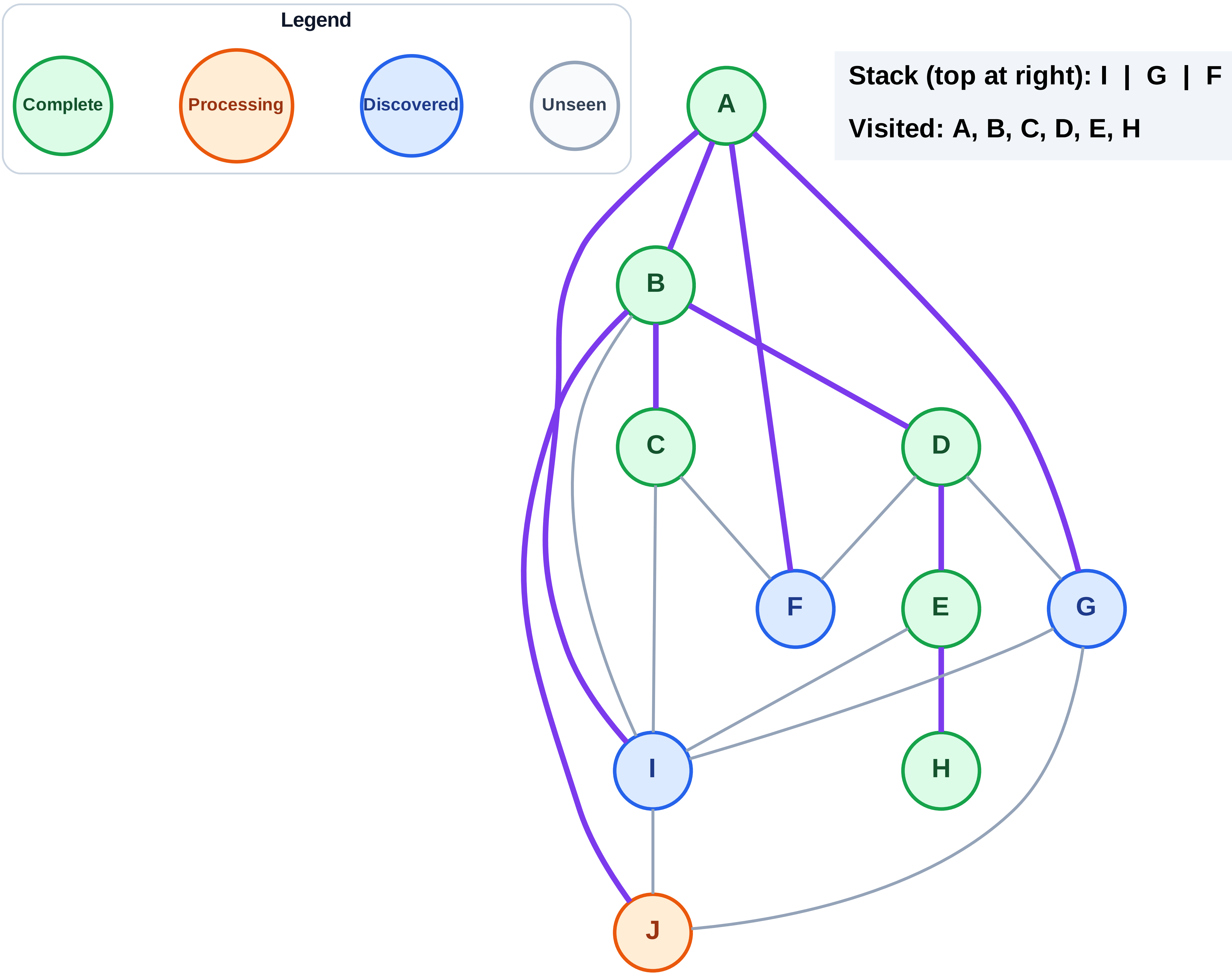
DFS Traversal

Step 13: No new neighbors from H



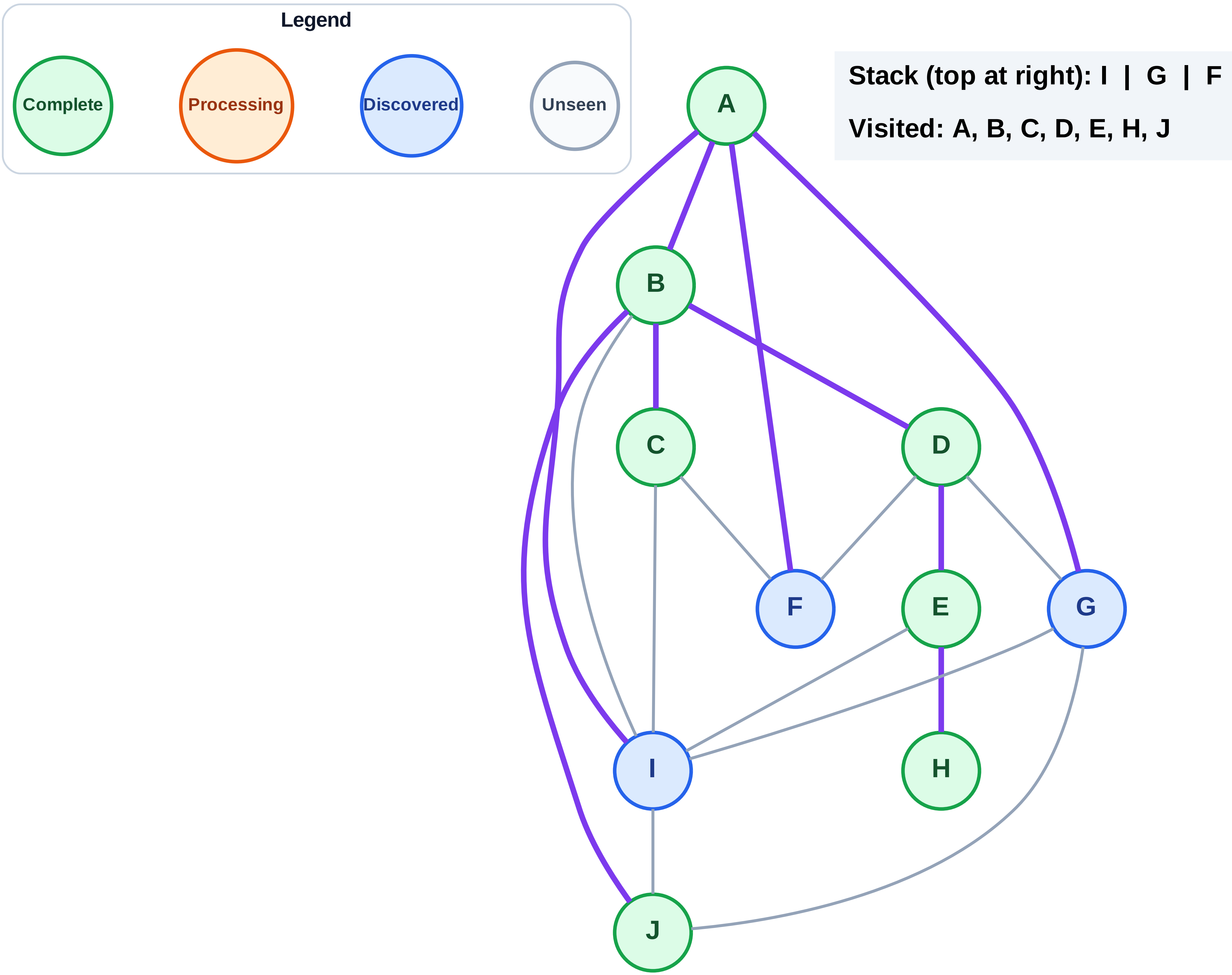
DFS Traversal

Step 14: Process node J



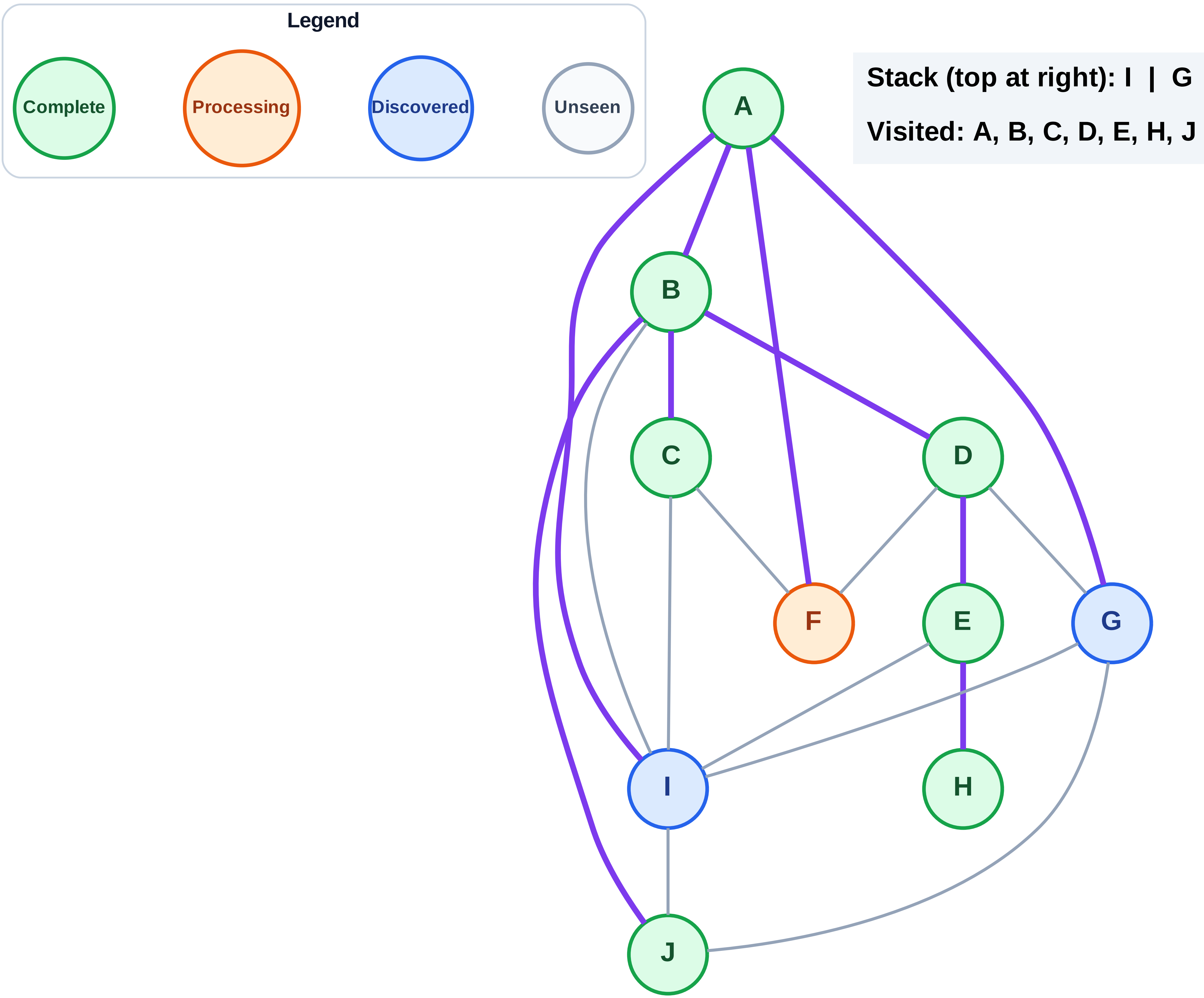
DFS Traversal

Step 15: No new neighbors from J



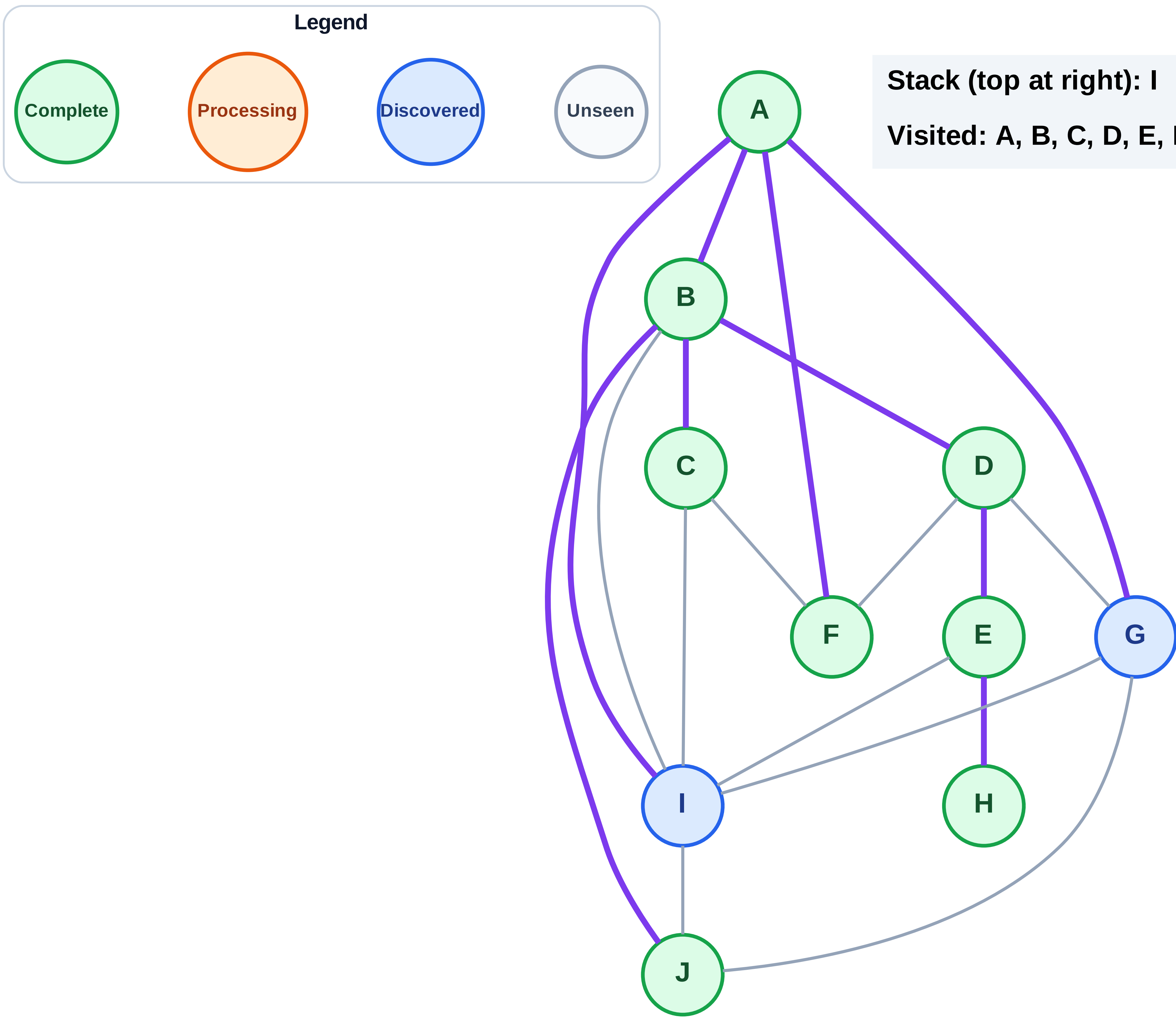
DFS Traversal

Step 16: Process node F



DFS Traversal

Step 17: No new neighbors from F

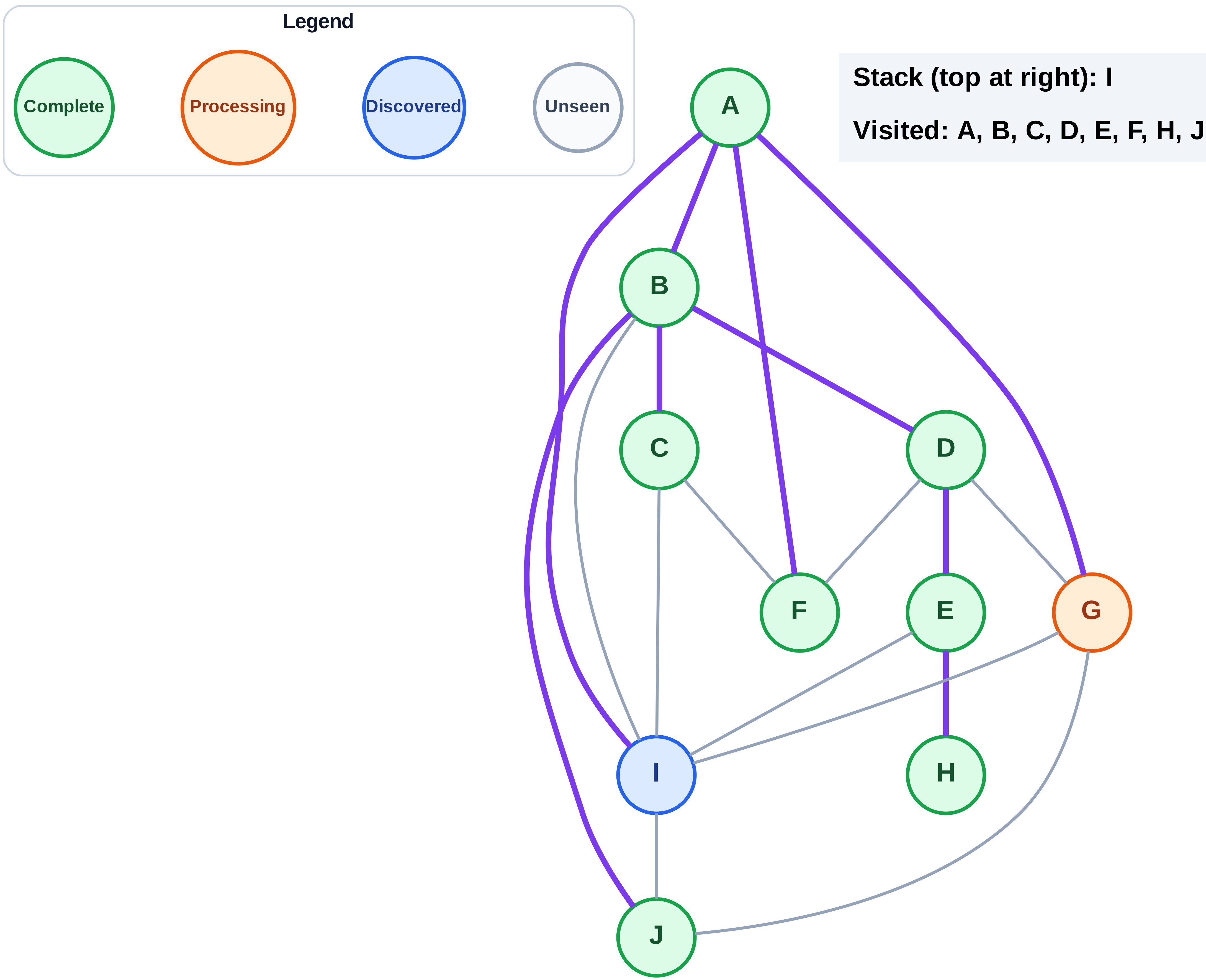


Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J

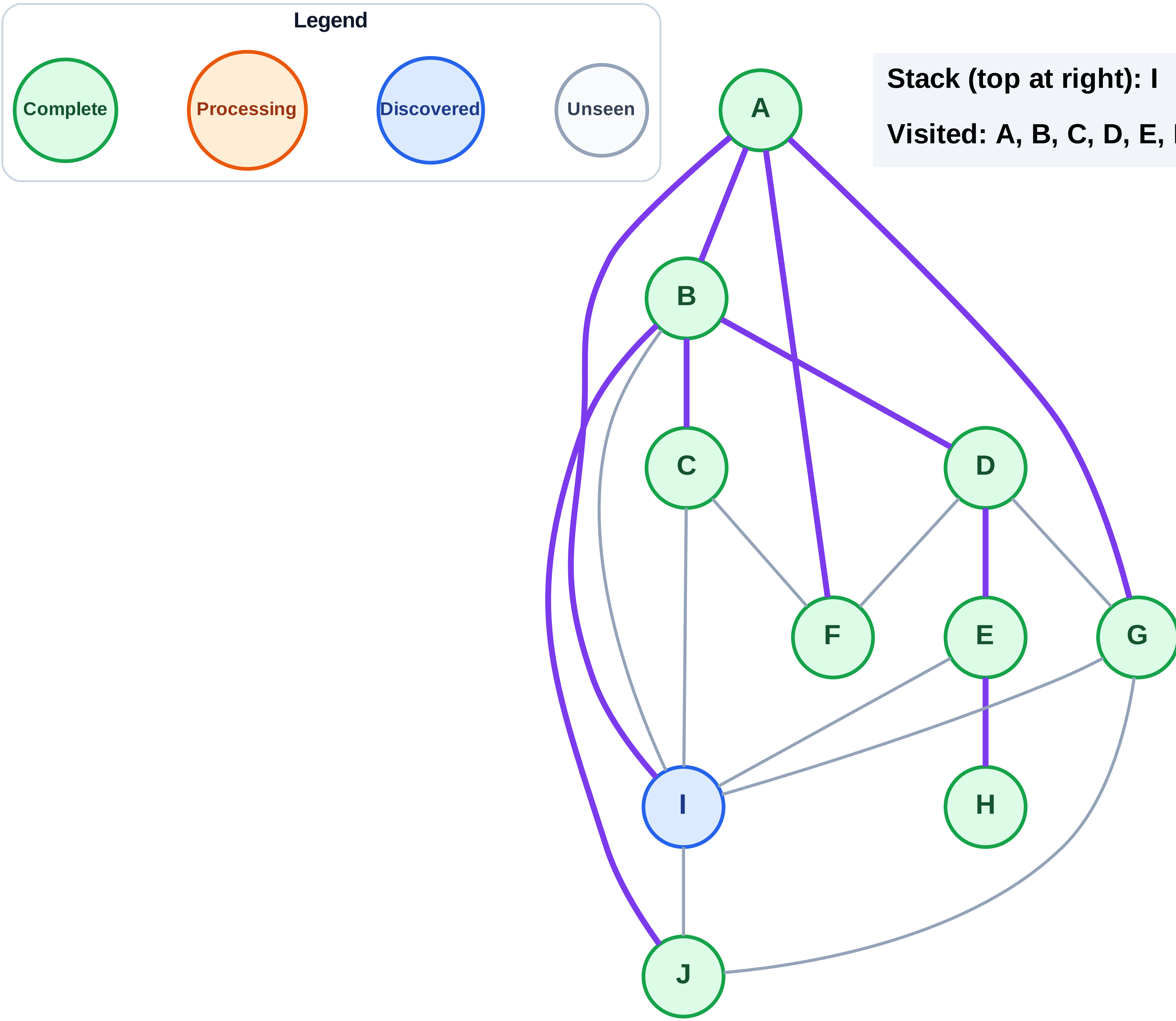
DFS Traversal

Step 18: Process node G



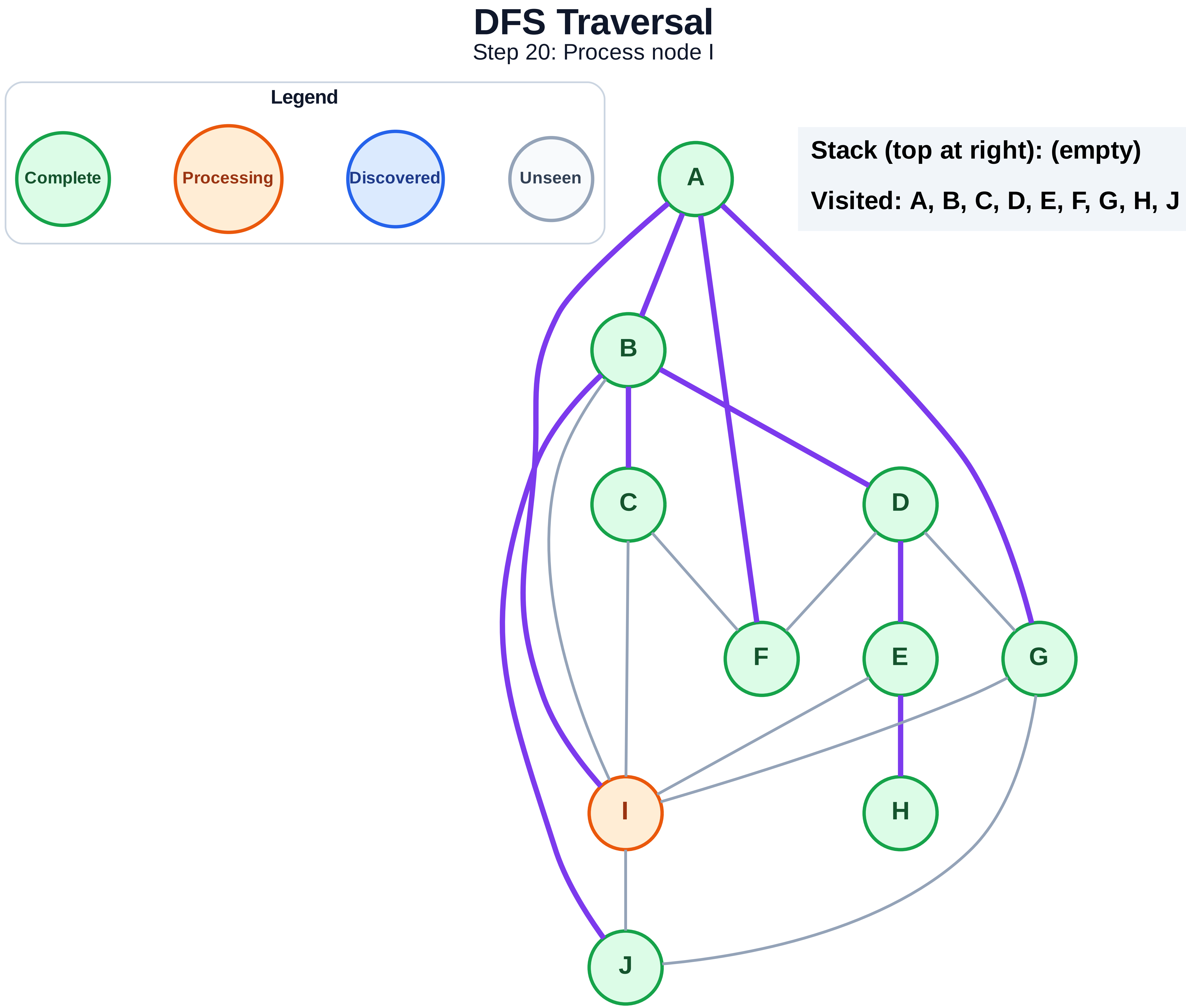
DFS Traversal

Step 19: No new neighbors from G



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J


DFS Traversal
Step 20: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J
A
B
C
D
F
E
G
I
H
J

DFS Traversal

Step 21: No new neighbors from I